O ahu Ōhi a Gradient — Full Report

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Table of contents

1 Introduction **3**

2 Plot 1: Sample Design and map **6**

3 Supl Plots: Data Preparation and Setup and initial visualiztion **6**

4 Plot 2: NMDS panel **12**

5 Statistical Analyses mainly focusing on ß-diversity **12**
5.1 Section 1: Insect-Community differences by Trees (Manana and Aiea Ridge) . . 12
5.2 Supl Plot: Beta-dispersion Test: Tree Type Group Variability 12
5.3 Section 2: Ōhi a – Community differences by Site 14
5.4 Supl Plot: Beta Diversity Decomposition 16
5.5 Section 3: Ōhi a – Community differences by elevation 19

6 Beta-diversity within between subsets **21**

7 Beta Diversity Decomposition based on Sorensen Family **23**

8 Geospatial Distance Decay and Mantel Test **28**

9 Supl Plot: Mantel-test and Distance Decay Plot **31**

10 Summary initial analyses **32**

11 Comparing Invasive Status Predictions vs Blast or Ibis **34**

12 Leke Transect Tree Comparison – Cleaned Code Chunk **34**

13 Plot 3: UpSet plot by BLAST status as example **34**

14 Supl. Plot : Heatmap of Confusion Matrix **37**

15 Elevation Bands and Invasion Status across all Oahu ridges **40**

16 Plot 4: Invasion among elevation bands **40**

17 Plot 5: UpSet plot of all five ridges **43**

18 Plot 6: Statistical differences of invasion between tree species on Aiea and Manana ridges **45**

19 Summary invasion status analysis **46**

20 Plot 7: Species Richness across all 'Ōhi'a samples from O'ahu, grouped by elevation and ridge **48**

21 Plot 8: Titan per-taxon change-points **63**

22 Summary TITAN **73**

Study Focus and Emerging Patterns This report investigates how arthropod community composition varies along elevation gradients on O ahu, with particular emphasis on the relative influence of site versus elevation, patterns of beta diversity turnover, and the distribution of invasive species across native and non-native host trees.

Emerging Hypotheses and Interpretations Beta diversity turnover increases with elevation. Community dissimilarity between sites is greater in high-elevation gradients than in low-elevation gradients, suggesting stronger environmental filtering or dispersal limitation at higher elevations.

Site effects dominate over elevation effects. Variation in species composition is explained more strongly by site identity than by elevation alone, indicating that local habitat characteristics, disturbance history, or microclimate may be primary drivers.

Community dispersion decreases at higher elevations. Contrary to expectations of greater stochasticity under harsher conditions, high-elevation sites show lower within-group dispersion compared to low- and mid-elevation sites.

Invasive species richness declines with elevation. The proportion of non-native taxa decreases along the elevation gradient, consistent with reduced propagule pressure or less suitable environmental conditions for invasives at higher elevations.

Host tree identity influences invasive species load. Across all elevations, Psidium cattleianum (strawberry guava) supports a higher proportion of invasive species compared to Acacia koa or Metrosideros polymorpha ('ōhi a). Communities on guava show the greatest variance in composition, particularly on Manana and Aiea ridges.

Species richness is higher at lower elevations. Lower-elevation sites exhibit greater total richness, possibly due to higher rates of invasion and greater habitat connectivity.

Site-specific anomalies warrant further investigation. The Poamoho trail displays community patterns that diverge markedly from other transects. Preliminary checks confirm the pattern is genuine, but the underlying cause remains unclear; its inclusion in final analyses will depend on interpretability.

1 Introduction

This document explores arthropod community composition across native and non-native tree species along elevational gradients using eDNA-based metabarcoding. We visualize these differences with NMDS and test their significance using PERMANOVA and beta-dispersion.

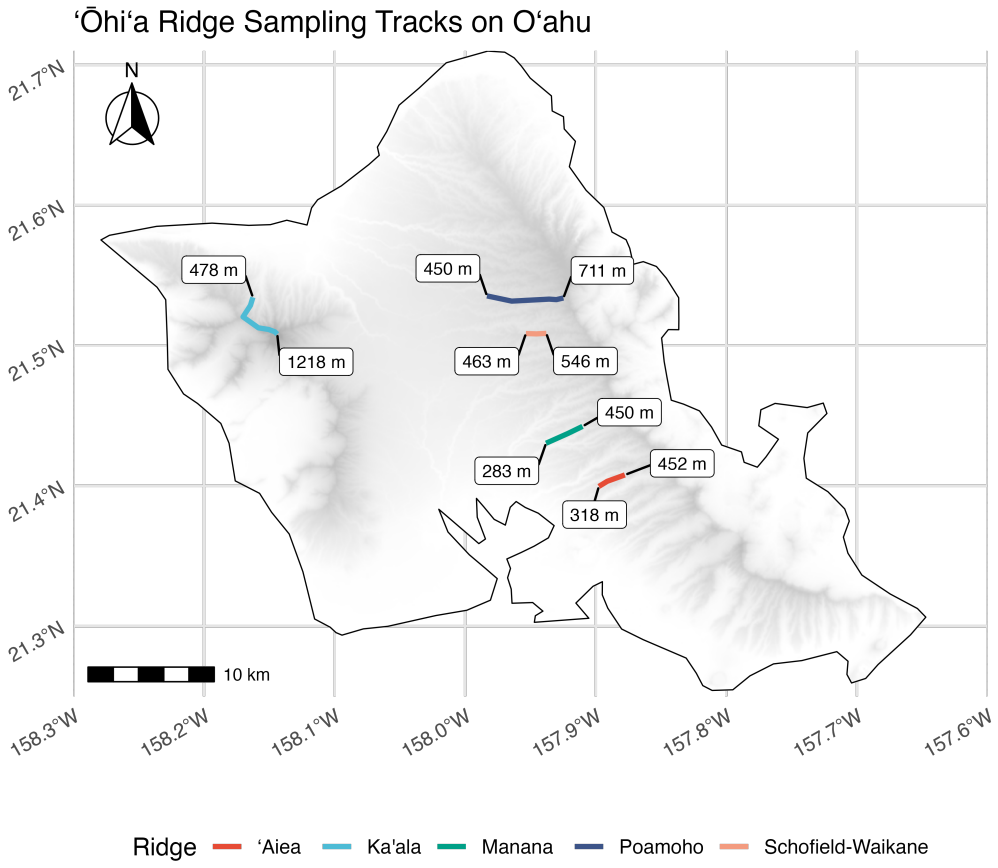
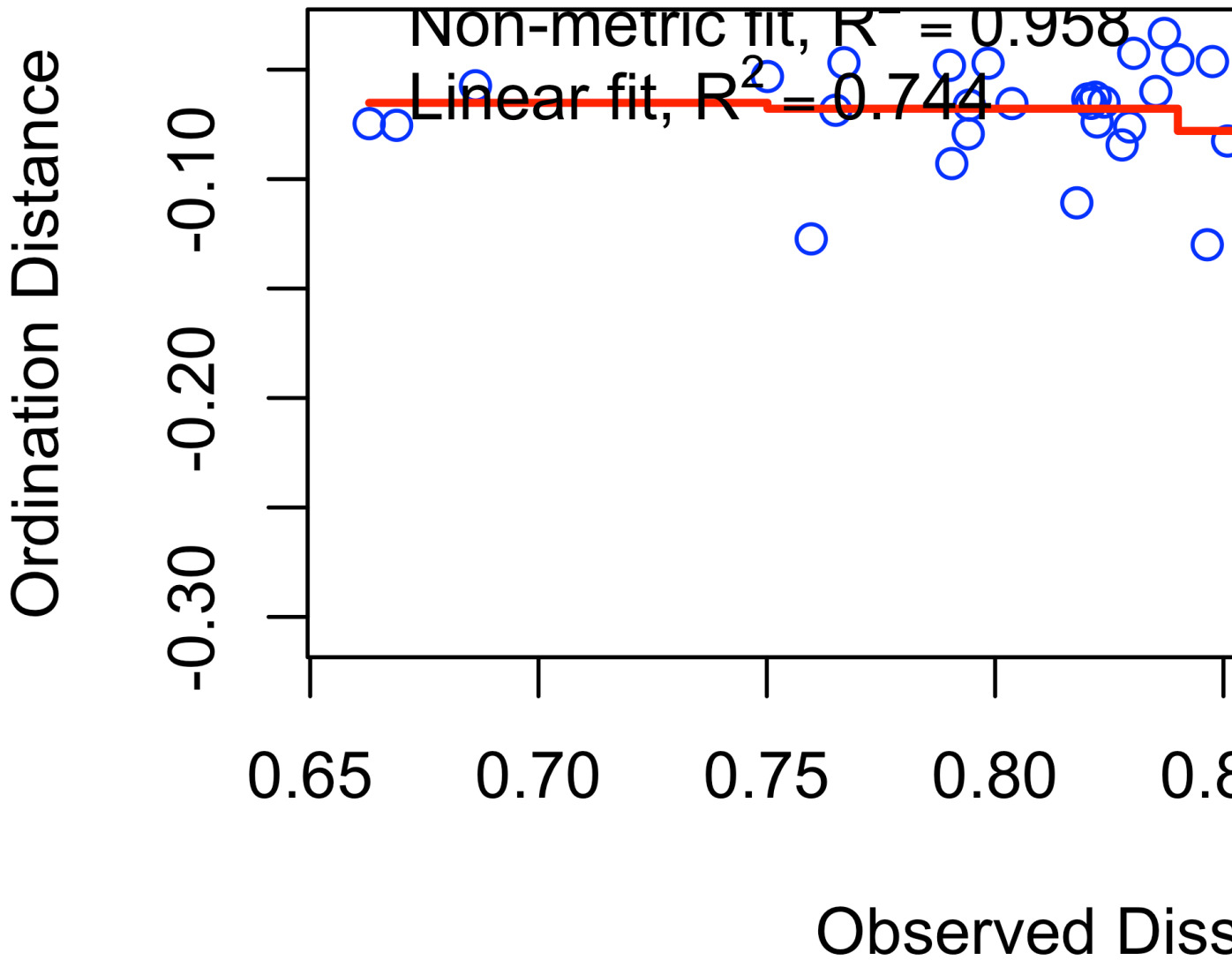
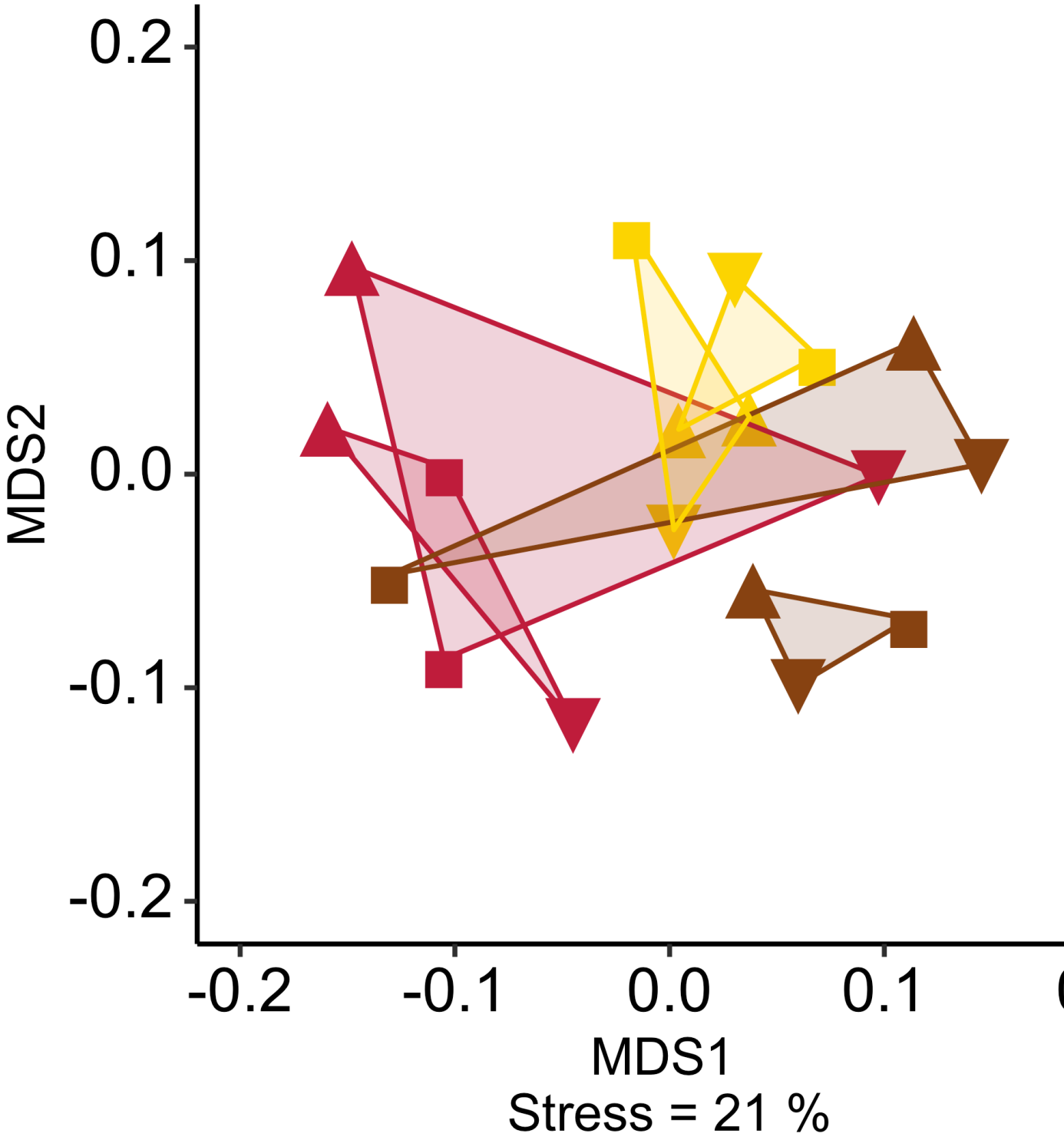


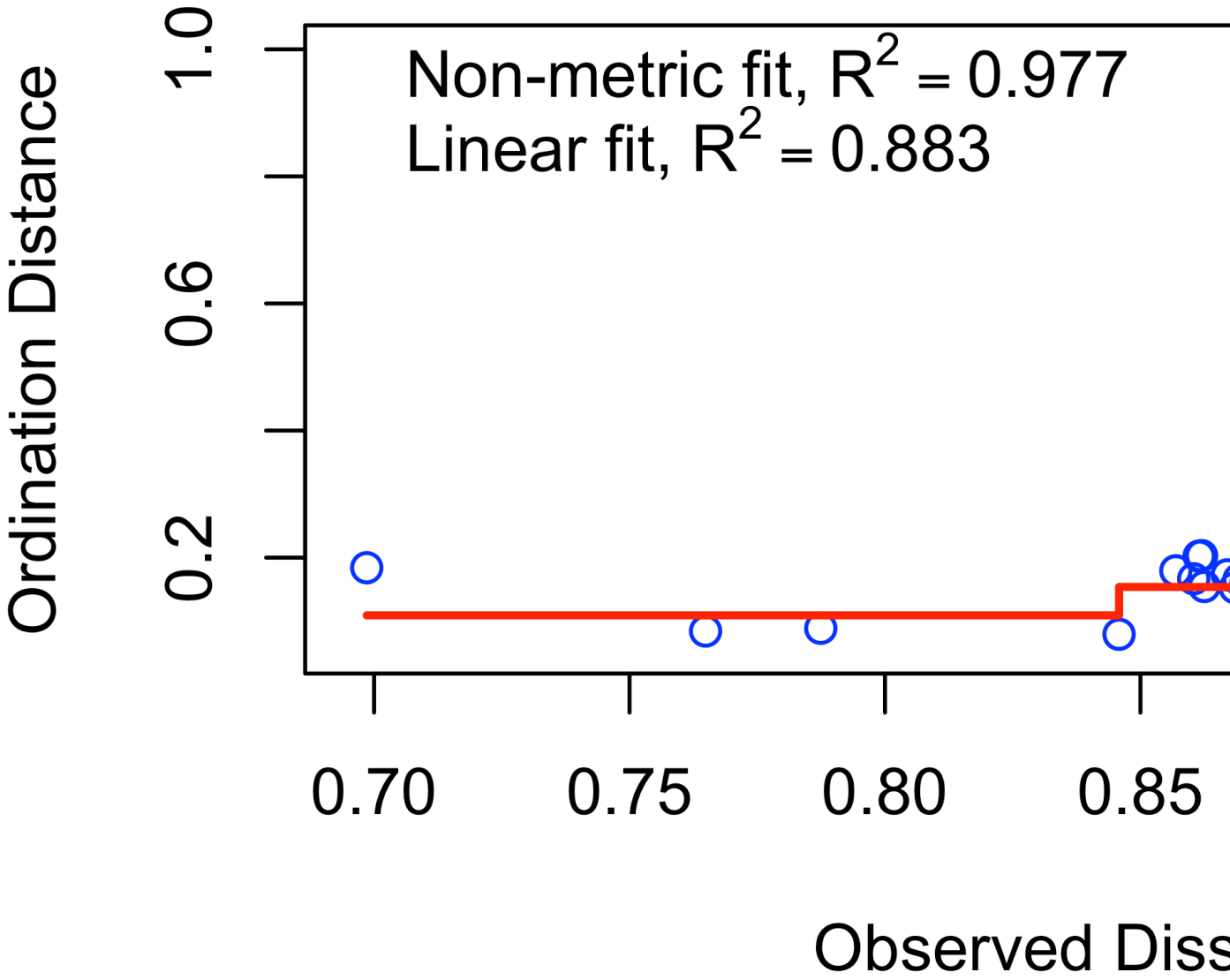
Figure 1: Figure 1.: Ridge sampling tracks across O ahu, Hawai i. Sampling tracks (colored by ridge) represent arthropod eDNA collections from Metrosideros polymorpha (ōhi a) along five elevation gradients. Base elevation was derived from a digital elevation model clipped to O ahu's outline, with values normalized to start at sea level. Labels indicate the minimum and maximum elevation of each ridge transect (in meters). Map elements include scale bar and north arrow for orientation. scale bar and north arrow for orientation.

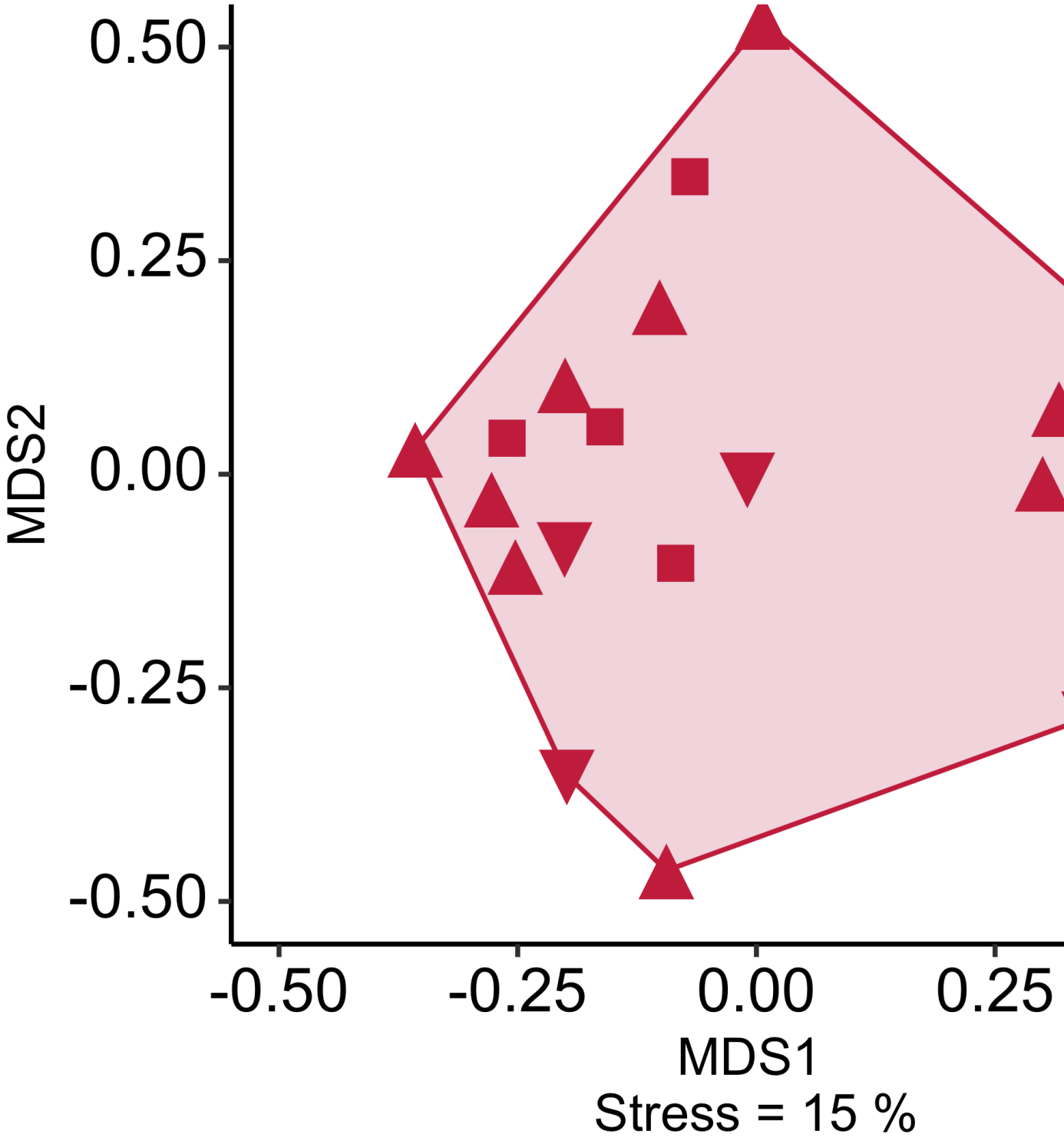
2 Plot 1: Sample Design and map

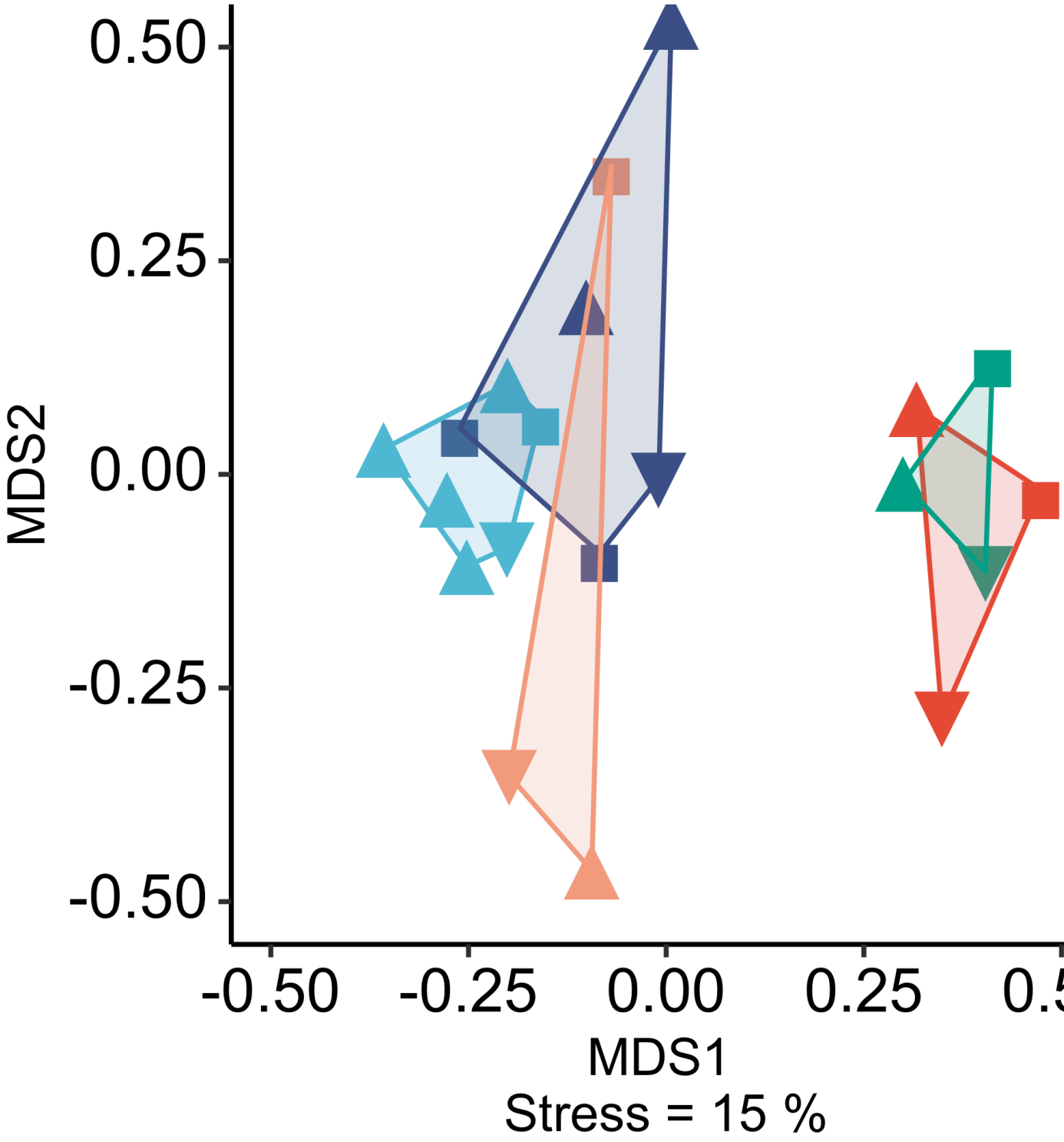
3 Supl Plots: Data Preparation and Setup and initial visualiztion

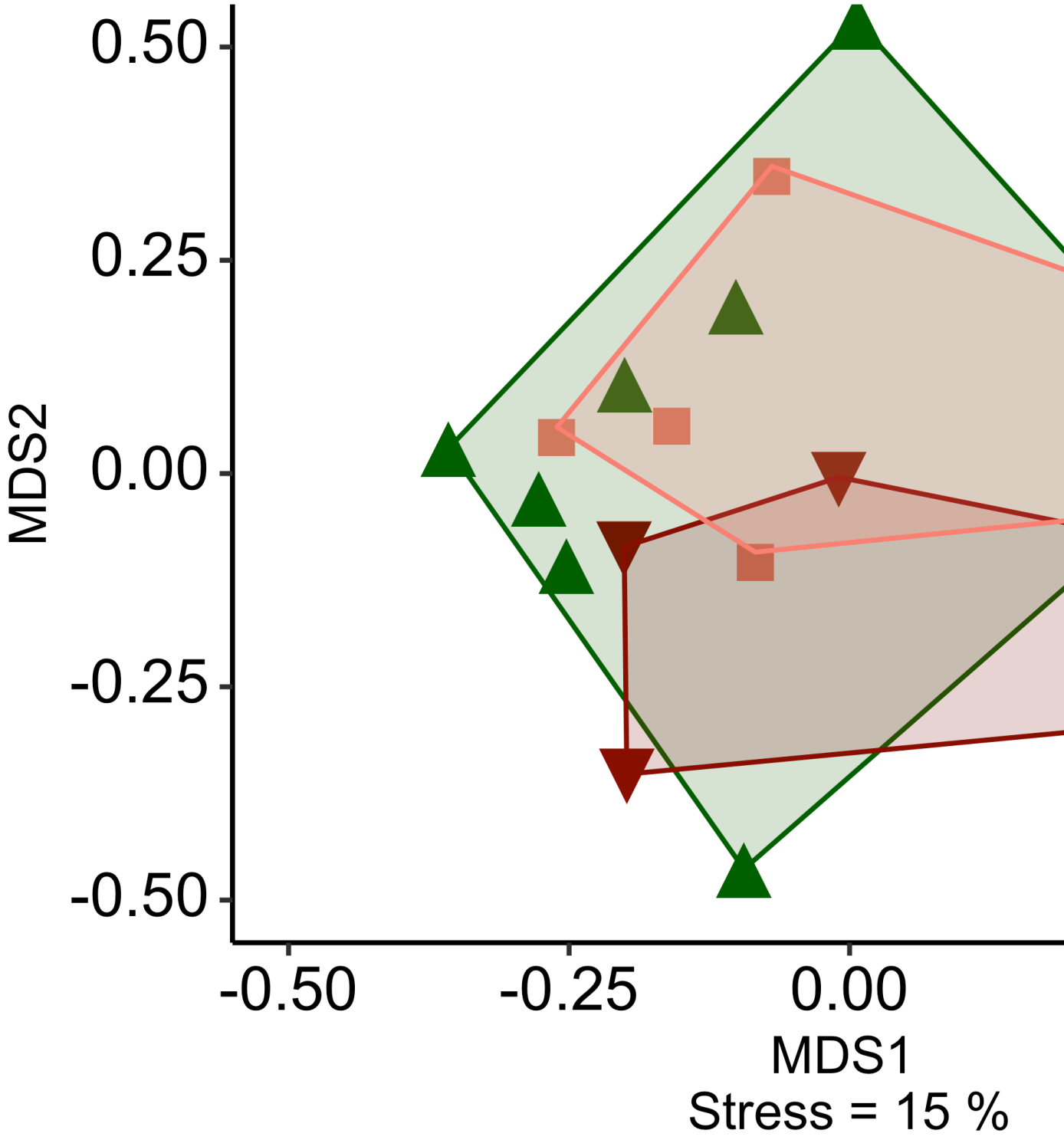



0.2
0.1
0.0
-0.1
-0.2
MDS2
-0.2
-0.1
0.0
0.1
MDS1
Stress = 21 %


Ordination Distance
1.0
0.6
0.2
Non-metric fit, R2 = 0.977
Linear fit, R2 = 0.883
0.70
0.75
0.80
0.85
Observed Diss


0.50
0.25
0.00
-0.25
-0.50
MDS2
-0.50
-0.25
0.00
0.25
MDS1
Stress = 15 %


0.50
0.25
0.00
-0.25
-0.50
MDS2
-0.50
-0.25
0.00
0.25
MDS1
Stress = 15 %


0.50
0.25
0.00
-0.25
-0.50
MDS2
-0.50
-0.25
0.00
MDS1
Stress = 15 %

4 Plot 2: NMDS panel

Figure X: Arthropod assemblages differ more strongly by site than elevation: Two-dimensional NMDS plots based on presence-absence data of arthropod communities from eDNA samples show that sampling site explains more variation in community composition than elevation.Top panel Comparison of the three most common trees across a small elevation gradient on Aiea & Manana Ridge. Stress value above 20% Middle panel: The same NMDS ordination, now grouped by sampling site; colored polygons highlight that samples from the same site cluster tightly despite differing in elevation. Each shape indicates the elevation level of the sample (triangles = high, inverted triangles = low, squares = mid). Stress value of 15% indicates a reliable two-dimensional representation of dissimilarities. Bottom panel: Samples grouped by elevation level (high, mid, low) across all sites; shaded polygons connect samples from the same elevation.

5 Statistical Analyses mainly focusing on ß-diversity

5.1 Section 1: Insect-Community differences by Trees (Manana and Aiea Ridge)

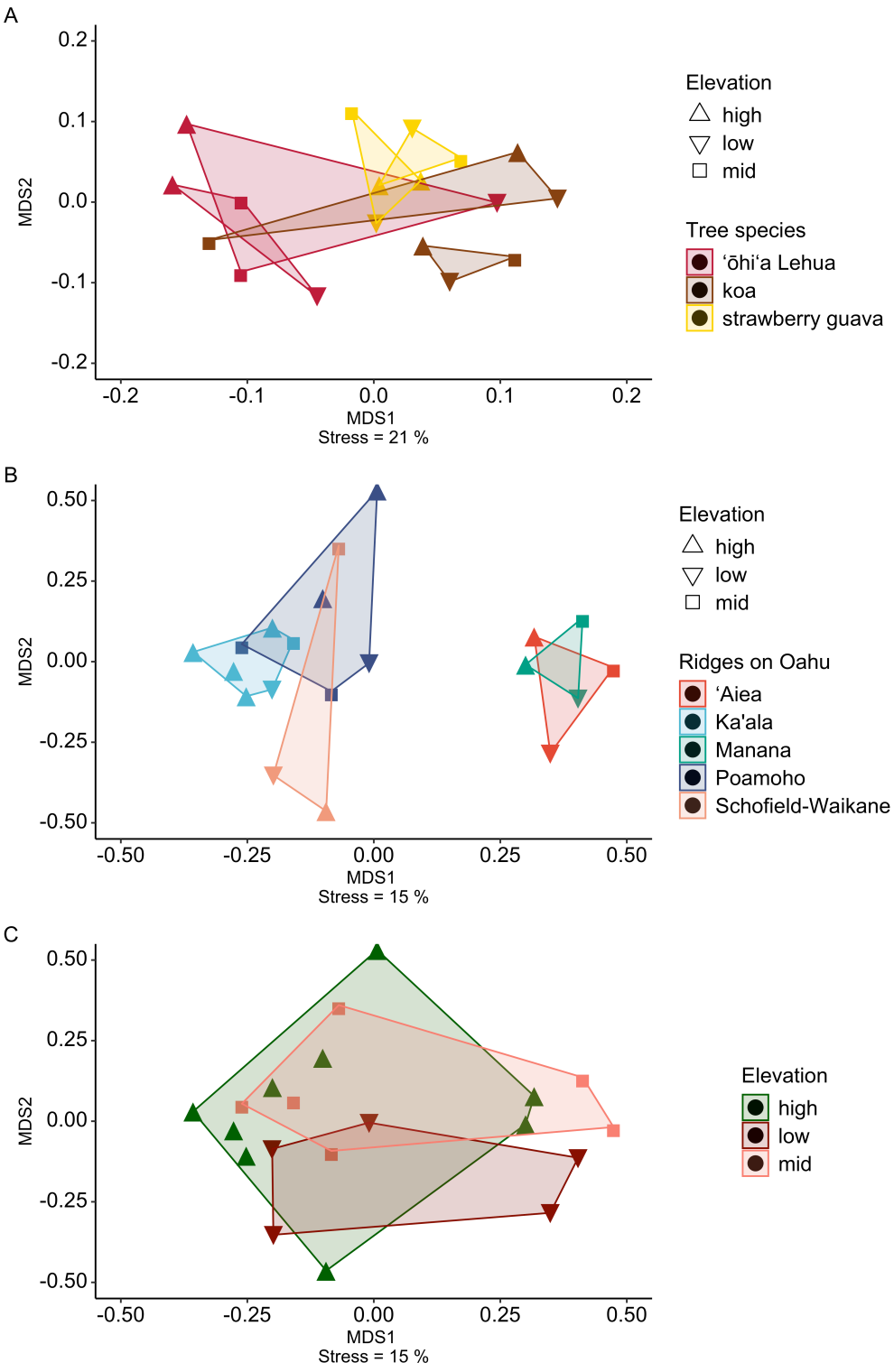
```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = bray_leke ~ commonname, data = meta_leke)
         Df SumOfSqs      R2      F Pr(>F)
Model     2   0.7383 0.09957 0.8294  0.893
Residual 15   6.6766 0.90043
Total    17   7.4149 1.00000
```

While the PERMANOVA test did not find significant differences in community composition between tree species (p = 0.895), the NMDS ordination suggests that samples from ōhi a, koa, and strawberry guava form visually distinct clusters. This pattern may indicate subtle differences in arthropod communities that are not statistically detectable at the current sample size. Also r squared value is very low, indicating that tree type does not explain much of the variation in community composition.

5.2 Supl Plot: Beta-dispersion Test: Tree Type Group Variability

```
Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999
```


A
0.2
0.1
0.0
-0.1
-0.2
MDS2
-0.2
-0.1
0.0
0.1
0.2
MDS1
Stress = 21 %
Elevation
high
low
mid
Tree species
‘ōhi‘a Lehua
koa
strawberry guava
B
0.50
0.25
0.00
-0.25
-0.50
MDS2
-0.50
-0.25
0.00
0.25
0.50
MDS1
Stress = 15 %
Elevation
high
low
mid
Ridges on Oahu
‘Aiea
Ka'ala
Manana
Poamoho
Schofield-Waikane
C
0.50
0.25
0.00
-0.25
-0.50
MDS2
-0.50
-0.25
0.00
0.25
0.50
MDS1
Stress = 15 %
Elevation
high
low
mid

```
Response: Distances
          Df    Sum Sq    Mean Sq      F N.Perm Pr(>F)
Groups     2 0.0008448 0.00042241 0.5116    999  0.591
Residuals 15 0.0123847 0.00082565
```

There is no significant difference in dispersion (beta diversity spread) among tree species ($p = 0.588$). This suggests the PERMANOVA assumption of equal group dispersions is not violated, which is good.

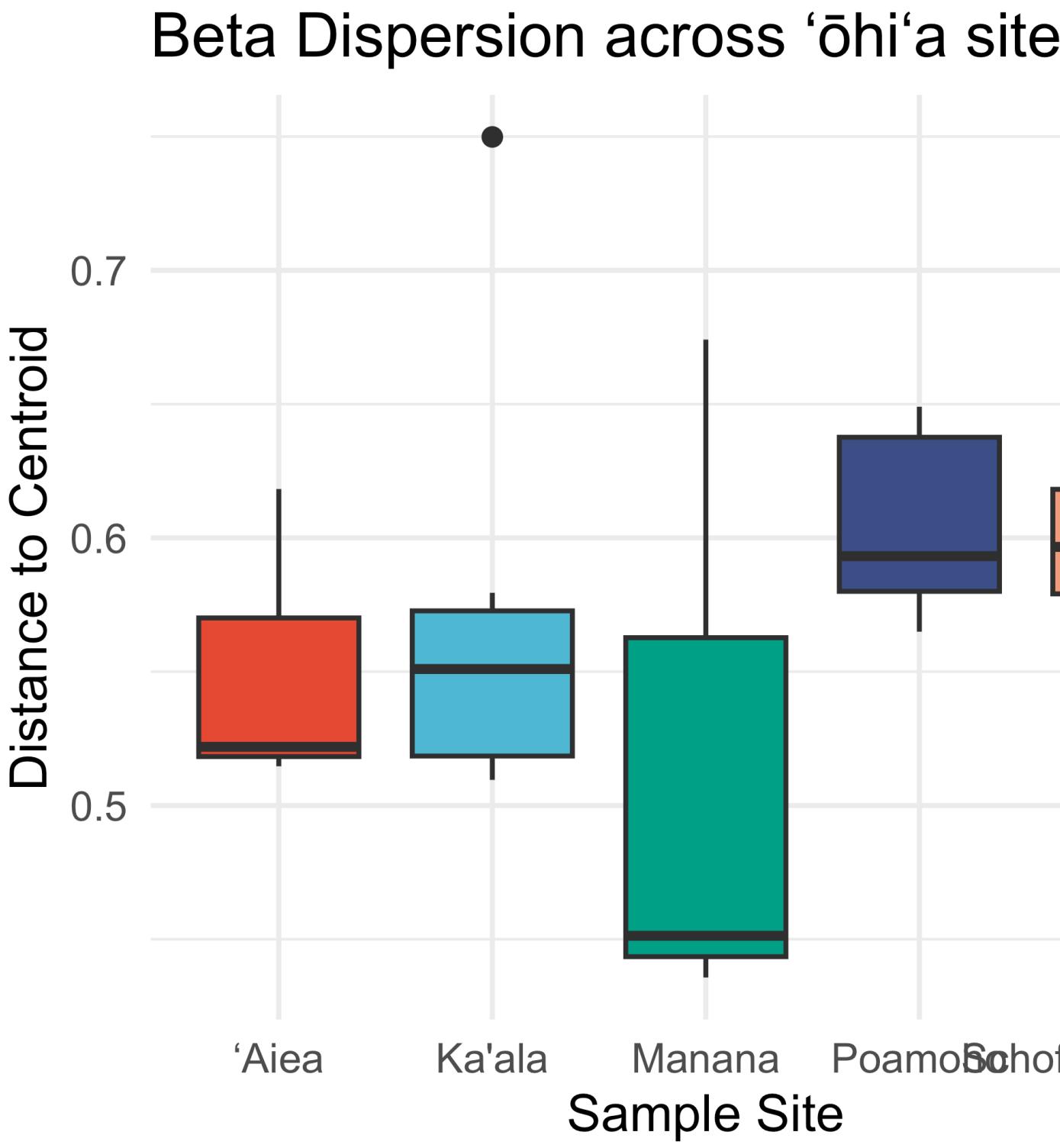
5.3 Section 2: Ōhi a – Community differences by Site

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = bray_ohia ~ sample_site, data = ohia_df)
         Df SumOfSqs      R2      F Pr(>F)
Model     6   3.4183 0.21033 1.2873  0.003 **
Residual 29  12.8339 0.78967
Total    35  16.2522 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

A PERMANOVA based on Bray-Curtis dissimilarities showed that arthropod community composition within ōhi a stands varied significantly by sampling site ($R^2 = 0.21$, $p = 0.002$), suggesting strong site-specific effects across the O ahu gradient.

5.4 Supl Plot: Beta Diversity Decomposition

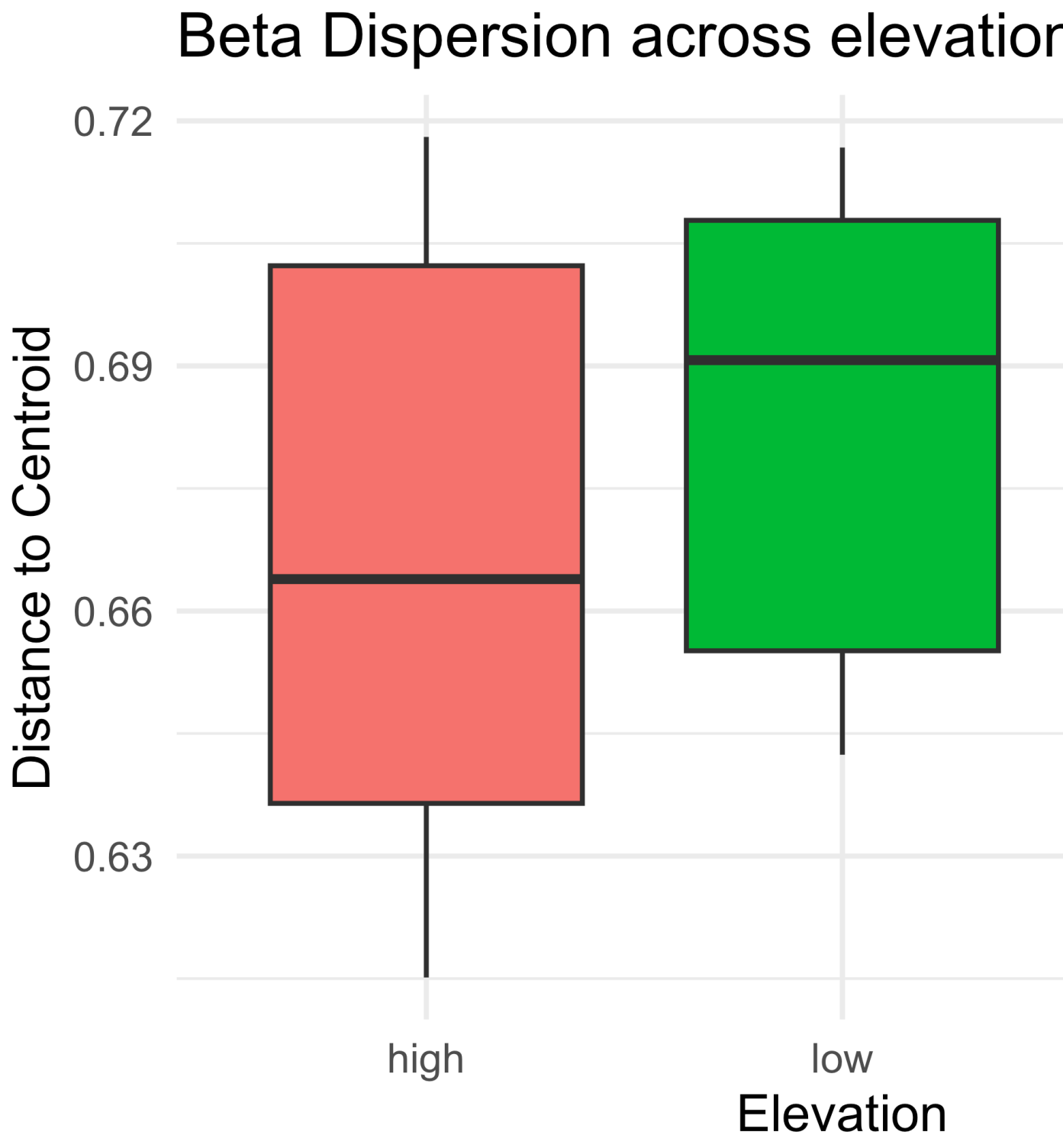


```
Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999

Response: Distances
          Df   Sum Sq   Mean Sq      F N.Perm Pr(>F)
Groups     6 0.041114 0.0068523 1.8714    999  0.119
Residuals 29 0.106185 0.0036615
```

Although community dispersion varied visually across sampling sites, a test for homogeneity of multivariate dispersions was not statistically significant ($F = 1.87$, $p = 0.119$). This supports the validity of the PERMANOVA result, indicating that observed differences in arthropod community composition across ōhi a sites are not driven by differences in within-site variability.

5.5 Section 3: Ōhi a – Community differences by elevation



```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = bray_ohia ~ elevation, data = ohia_df)
         Df SumOfSqs      R2      F Pr(>F)
Model     2   0.9365 0.05762 1.0089  0.388
Residual 33  15.3157 0.94238
Total    35  16.2522 1.00000


Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999

Response: Distances
          Df   Sum Sq   Mean Sq      F N.Perm Pr(>F)
Groups     2 0.011088 0.0055442 5.1781    999  0.015 *
Residuals 33 0.035333 0.0010707
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We found no significant differences in arthropod community composition across elevations within ōhi a stands (PERMANOVA: $R^2 = 0.058$, $p = 0.401$). However, a test for homogeneity of dispersions revealed that beta diversity varied significantly across elevations ($F = 5.18$, $p = 0.013$), indicating that community consistency differs with elevation despite no shift in centroids.

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = bray_ohia_oahu ~ sample_site + elevation, data = ohia_oahu_df)
         Df SumOfSqs      R2     F Pr(>F)
Model     6   3.2667 0.33582 1.264  0.001 ***
Residual 15   6.4607 0.66418
Total    21   9.7274 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

A combined PERMANOVA model revealed that site and elevation together significantly structured arthropod community composition within ōhi a stands across O ahu ($R^2 = 0.34$, $p =$

0.001). These variables jointly explained ~34% of the observed variation, indicating that spatial and elevational differences play a strong role in shaping arthropod assemblages within this native forest type.

6 Beta-diversity within between subsets

```
    Kruskal-Wallis rank sum test

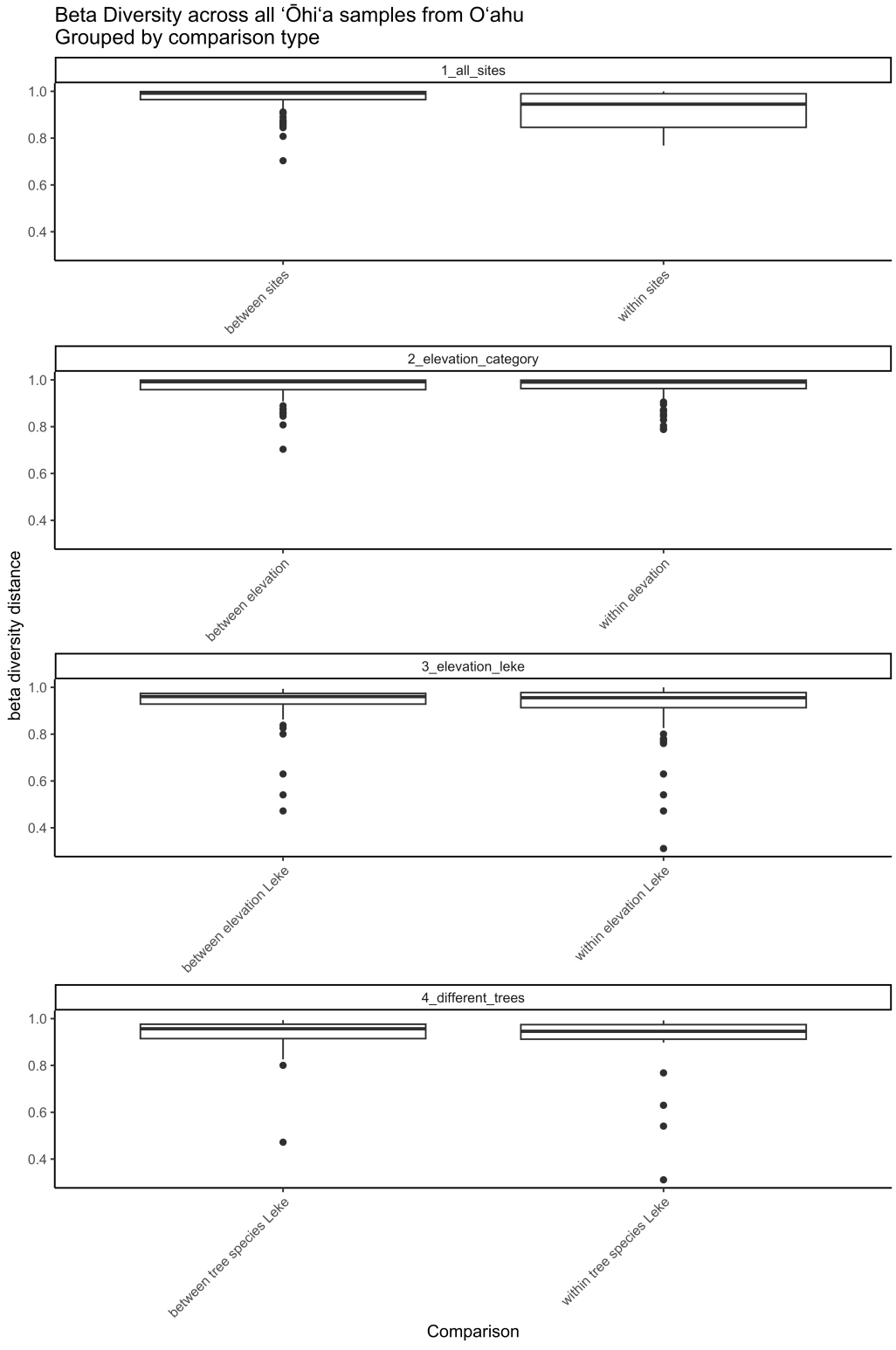
data:  distance by comparison
Kruskal-Wallis chi-squared = 195.47, df = 7, p-value < 2.2e-16


    Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  beta_comp$distance and beta_comp$comparison

                         between elevation between elevation Leke
between elevation Leke   3.7e-07          -
between sites            0.92             8.1e-09
between tree species Leke 2.9e-07         0.92
within elevation         0.76             1.4e-07
within elevation Leke    1.2e-14          0.92
within sites             2.9e-07          0.98
within tree species Leke 1.9e-10          0.60
                         between sites between tree species Leke
between elevation Leke   -             -
between sites            -             -
between tree species Leke 8.1e-09      -
within elevation         0.60          2.8e-07
within elevation Leke    < 2e-16       0.98
within sites             8.1e-09       0.92
within tree species Leke 1.7e-13       0.76
                         within elevation within elevation Leke within sites
between elevation Leke   -               -                     -
between sites            -               -                     -
between tree species Leke -              -                     -
within elevation         -               -                     -
within elevation Leke    3.9e-16         -                     -
within sites             6.9e-07         0.92                  -
within tree species Leke 6.9e-11         0.61                  0.92

P value adjustment method: fdr
```


Beta Diversity across all ‘Ōhi‘a samples from O‘ahu
Grouped by comparison type
1_all_sites
2_elevation_category
3_elevation_leke
4_different_trees
beta diversity distance
1.0
0.8
0.6
0.4
between sites
within sites
between elevation
within elevation
between elevation Leke
within elevation Leke
between tree species Leke
within tree species Leke
Comparison

Pairwise Wilcoxon tests comparing Bray-Curtis dissimilarities revealed significantly greater beta diversity between elevations, tree species, and sites than within these groups (FDR-corrected $p < 0.001$ for most contrasts). Notably, arthropod communities differed significantly between tree species where all three co-occurred, and between elevation bands both within and across sites. Within-site and within-elevation comparisons consistently showed lower dissimilarity, indicating stronger community cohesion at those scales.

7 Beta Diversity Decomposition based on Sorensen Family

We want to expose the components of beta diversity, specifically turnover and nestedness, using the Sorensen index. This will help us understand how much of the dissimilarity is due to species turnover versus nestedness.

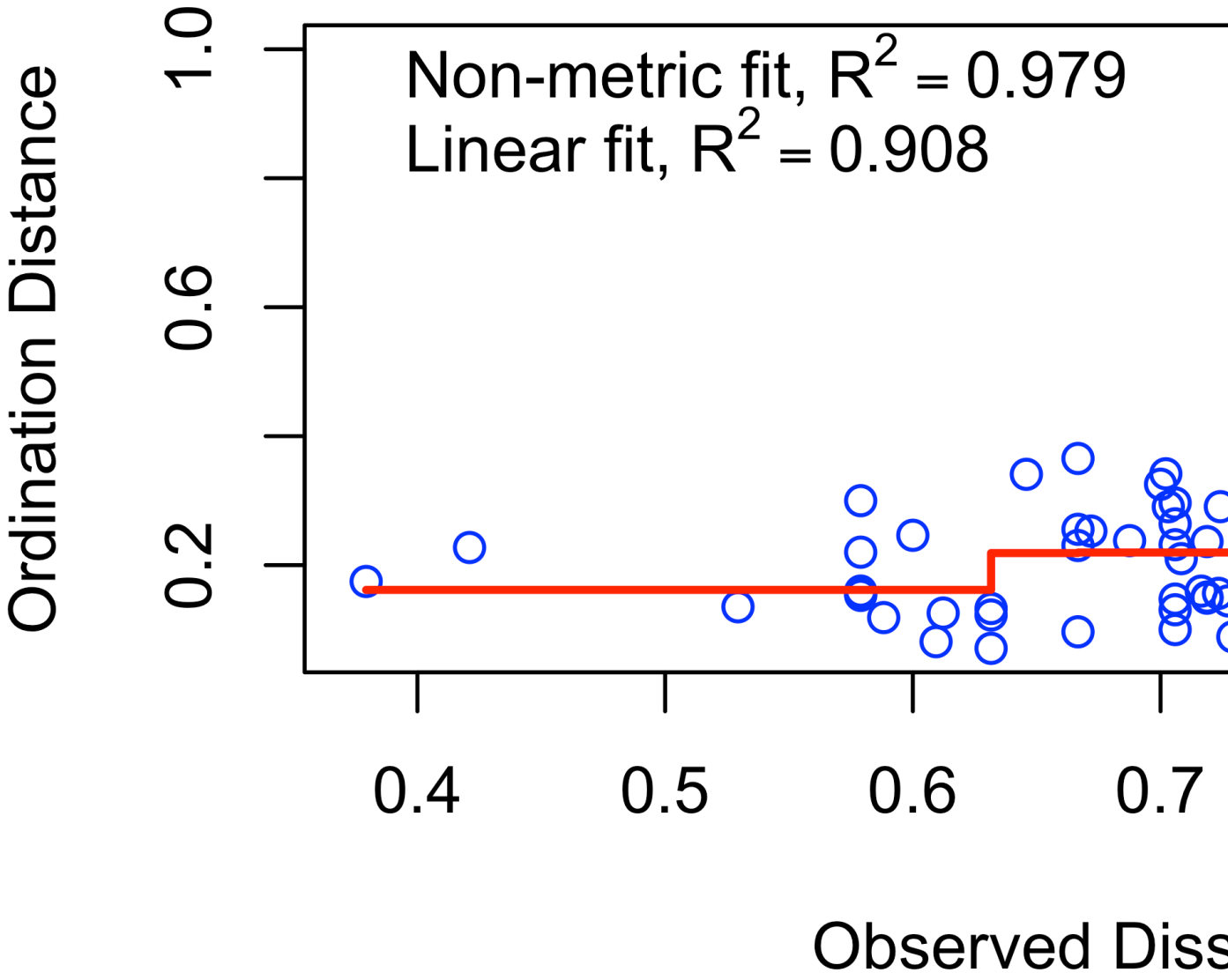
-Are differences between communities driven by species replacement (turnover)?

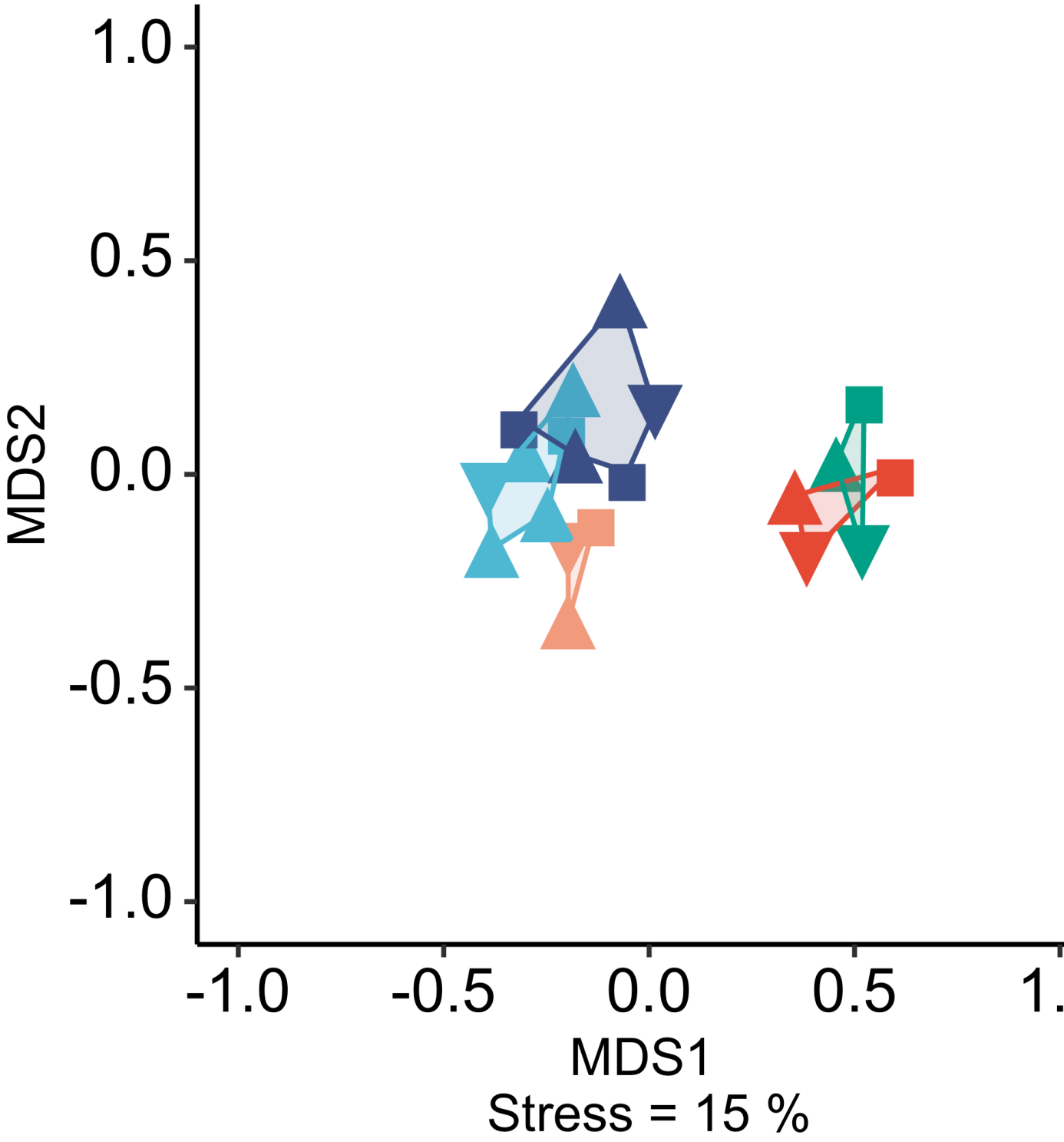
-Or by nestedness (e.g. lowland sites losing taxa)?

-Which sites are more compositionally heterogeneous (high within-site turnover)?

This goes beyond Bray-Curtis, offering more ecological insight.

NMDS based on beta.sim will show us the turnover component of beta diversity, which is the species replacement component. This will help us understand how much of the dissimilarity is due to species turnover versus nestedness.


Non-metric fit, $R^2 = 0.979$
Linear fit, $R^2 = 0.908$
Ordination Distance
0.2
0.6
1.0
0.4
0.5
0.6
0.7
Observed Diss


1.0
0.5
0.0
-0.5
-1.0
MDS2
-1.0
-0.5
0.0
0.5
1.
MDS1
Stress = 15 %

```
Call:
protest(X = nmdsohia, Y = nmds_ohia_turn)

Number of objects: 20    Number of dimensions: 2

Procrustes sum of squares:
 0.2084693
Procrustes root mean squared error:
 0.1020954
Quantiles of Procrustes errors:
       Min         1Q     Median         3Q        Max
0.01654698 0.03764614 0.07503118 0.10889295 0.29041273

Rotation matrix:
           [,1]        [,2]
[1,] 0.99994361 -0.01061937
[2,] 0.01061937  0.99994361

Translation of averages:
             [,1]         [,2]
[1,] 7.432452e-18 2.171364e-18

Scaling of target:
[1] 0.8896801
```

MDS ordinations based on Bray–Curtis and Sørensen turnover dissimilarities both revealed strong ridge-level structuring of arthropod communities. While the Bray–Curtis NMDS (Stress = 15%) yielded a slightly better fit than the Sørensen-based NMDS (Stress = 17%), both metrics showed consistent clustering. The Sørensen ordination emphasizes species turnover alone, confirming that site-level differences are driven not only by shifts in abundance but also by distinct species pools. The Bray–Curtis plot further suggests that relative abundance patterns reinforce community structure, particularly within elevation bands.

A Procrustes analysis comparing the Bray–Curtis and Sørensen turnover NMDS ordinations revealed a high degree of congruence (Procrustes sum of squares = 0.208, RMSE = 0.102), with minimal rotation or scaling required. This confirms that site-level community differences are primarily driven by species turnover, and that patterns are consistent across presence–absence and abundance-based metrics

```
# A tibble: 5 x 2
  sample_site       mean_turnover
  <chr>                     <dbl>
1 Ka'ala                    0.763
```

```
2 Manana                    0.636
3 Poamoho                   0.730
4 Schofield-Waikane         0.807
5  Aiea                    0.763

Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = beta_part$beta.sim ~ sample_site + elevation, data = ohia_meta)
         Df SumOfSqs      R2      F Pr(>F)
Model     6   3.1945 0.46492 1.8826  0.001 ***
Residual 13   3.6766 0.53508
Total    19   6.8711 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999

Response: Distances
          Df   Sum Sq  Mean Sq      F N.Perm Pr(>F)
Groups     4 0.040677 0.010169 0.7795    999  0.566
Residuals 15 0.195683 0.013045


    Kruskal-Wallis rank sum test

data:  mean_turnover by sample_site
Kruskal-Wallis chi-squared = 4, df = 4, p-value = 0.406


    Pairwise comparisons using Wilcoxon rank sum exact test

data:  site_turnover$mean_turnover and site_turnover$sample_site

                  Aiea Ka'ala Manana Poamoho
Ka'ala            1    -      -      -
Manana            1    1      -      -
Poamoho           1    1      1      -
Schofield-Waikane 1    1      1      1

P value adjustment method: fdr
```

Decomposition of beta diversity showed that community dissimilarity among ōhi a stands was primarily driven by species turnover, rather than nestedness. A PERMANOVA based on turnover distances revealed significant effects of site and elevation ($R^2 = 0.38$, $p = 0.001$), while a dispersion test confirmed homogeneity of within-group variance ($p = 0.679$). Mean within-site turnover ranged from 0.68 to 0.81, with no significant differences among sites (Kruskal–Wallis $p = 0.406$), indicating consistently high compositional variability across the landscape.

8 Geospatial Distance Decay and Mantel Test

```
Call:
lm(formula = bray_vals ~ geo_vals)

Residuals:
     Min       1Q   Median       3Q      Max
-0.62590 -0.01083  0.01616  0.03756  0.06456

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 9.331e-01  4.735e-03   197.0  < 2e-16 ***
geo_vals    1.881e-06  2.767e-07     6.8 2.86e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.06514 on 526 degrees of freedom
Multiple R-squared:  0.0808,    Adjusted R-squared:  0.07905
F-statistic: 46.24 on 1 and 526 DF,  p-value: 2.856e-11
```

Community dissimilarity increased significantly with geographic distance ($R^2 = 0.081$, $p < 0.0001$). While the effect size is modest, this supports a weak but significant distance-decay pattern of arthropod communities across ridges and elevations. Even across small spatial scales, localized turnover contributes to beta diversity.

```
Mantel statistic based on Pearson's product-moment correlation

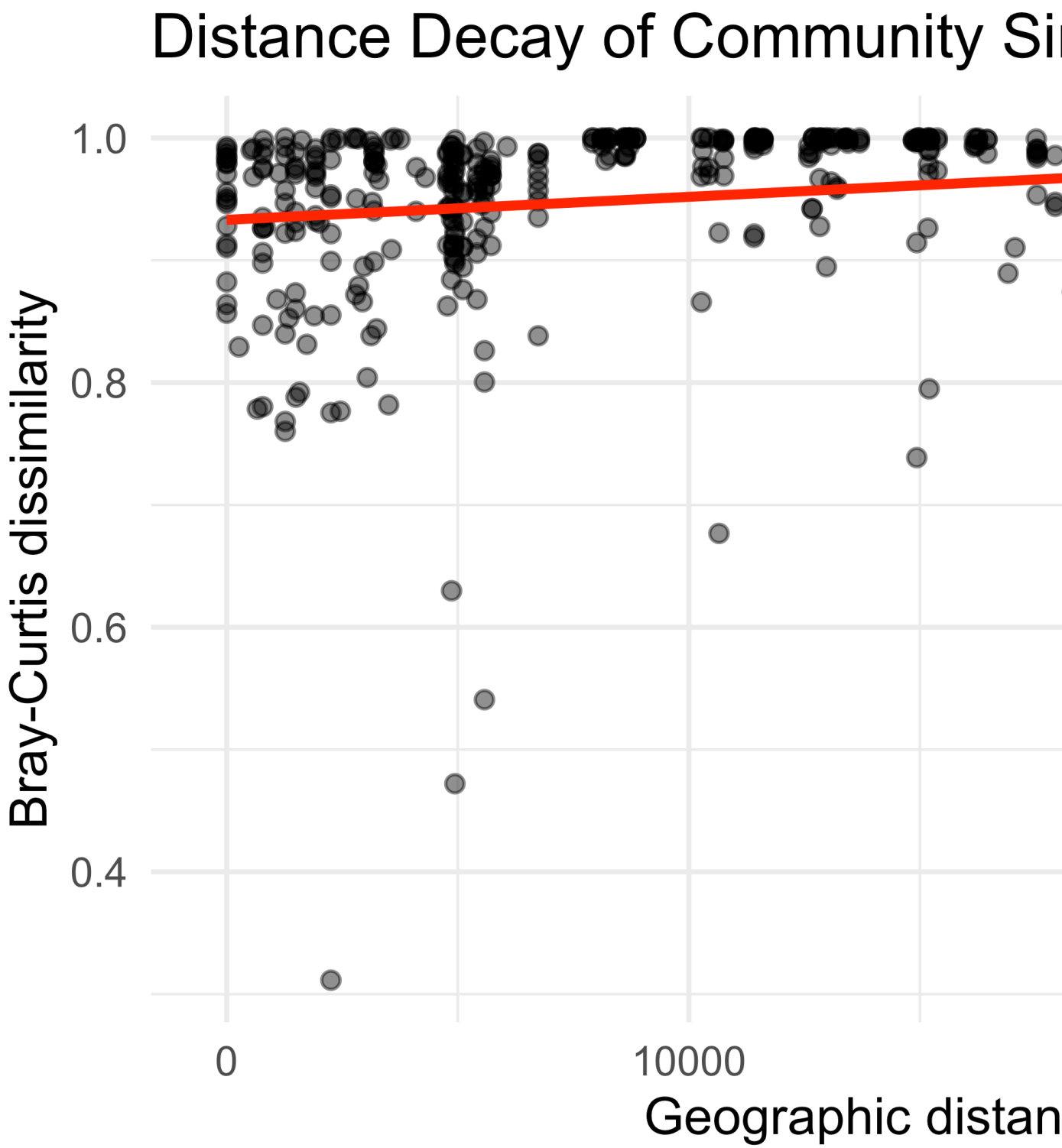
Call:
mantel(xdis = as.dist(geo_dist), ydis = bray_dist, method = "pearson",      permutations = 9

Mantel statistic r: 0.2842
      Significance: 1e-04

Upper quantiles of permutations (null model):
```

```
   90%    95%  97.5%    99%
0.0768 0.0993 0.1186 0.1406
Permutation: free
Number of permutations: 9999
```

9 Supl Plot: Mantel-test and Distance Decay Plot



A Mantel test revealed a significant correlation between Bray-Curtis dissimilarity and geographic distance ($r = 0.284$, $p = 0.0001$), indicating a moderate but statistically robust distance-decay pattern in arthropod community composition across sampling sites.

This suggests that geographic separation contributes to community turnover, even after accounting for local habitat effects. The result aligns with the idea that dispersal limitation or historical biogeography plays a role in shaping beta diversity along Hawaiian forest gradients.

10 Summary initial analyses

This report demonstrates that arthropod communities across Hawaiian forest gradients are shaped by a combination of spatial structure, elevational zonation, and forest composition. Using eDNA-based metabarcoding and a suite of multivariate analyses, we find:

Community composition in ōhi a forests varies significantly across sites (PERMANOVA $R^2 = 0.21$, $p = 0.002$) and is further influenced by elevation (combined $R^2 = 0.34$, $p = 0.001$).

Tree species identity, when tested in co-occurring sites (e.g., Manana and Aiea), does not explain significant variation in community structure ($p = 0.895$), though NMDS ordinations suggest subtle clustering.

Beta diversity decomposition reveals that community differences are primarily driven by species turnover rather than nestedness, with turnover significantly structured by site and elevation ($R^2 = 0.38$, $p = 0.001$).

Beta-dispersion patterns support consistent within-site heterogeneity and do not confound PERMANOVA results.

Distance–decay patterns indicate a weak but significant effect of geographic distance on community dissimilarity (Mantel $r = 0.284$, $p = 0.0001$), suggesting that even over short distances, dispersal limitation or spatial structuring contributes to beta diversity.

Together, these findings highlight the strong role of site-specific conditions and elevation in structuring arthropod communities in Hawaiian forests, while tree species identity plays a secondary role. The integration of turnover-based decomposition and spatial statistics adds ecological depth and supports the use of eDNA as a scalable tool for arthropod biodiversity monitoring across complex landscapes.

Summary of Invasive Status Predictions

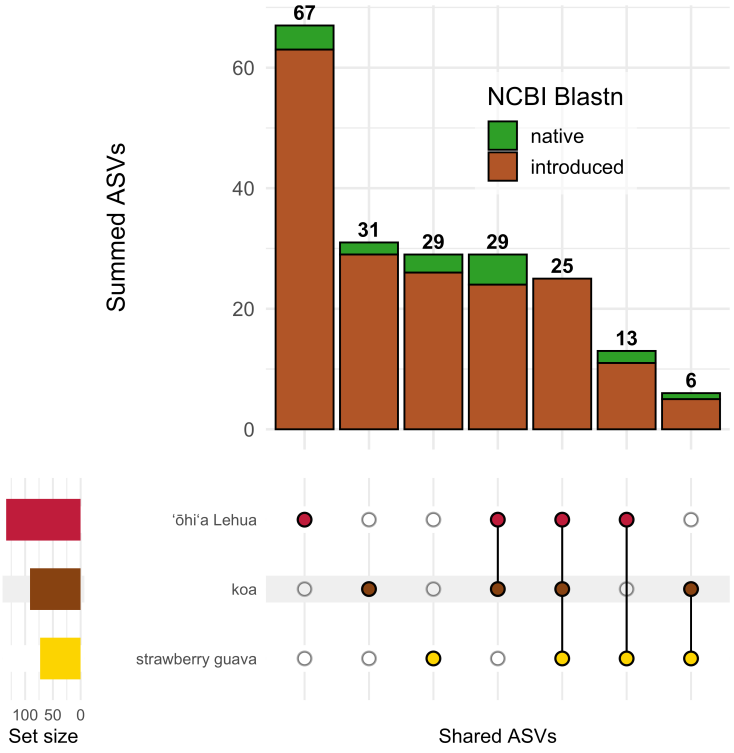
Prediction Method	Invasive Status		
	introduced	native	No information available
Blast	763	64	24
Ibis	433	22	396
NiClassify	394	457	0

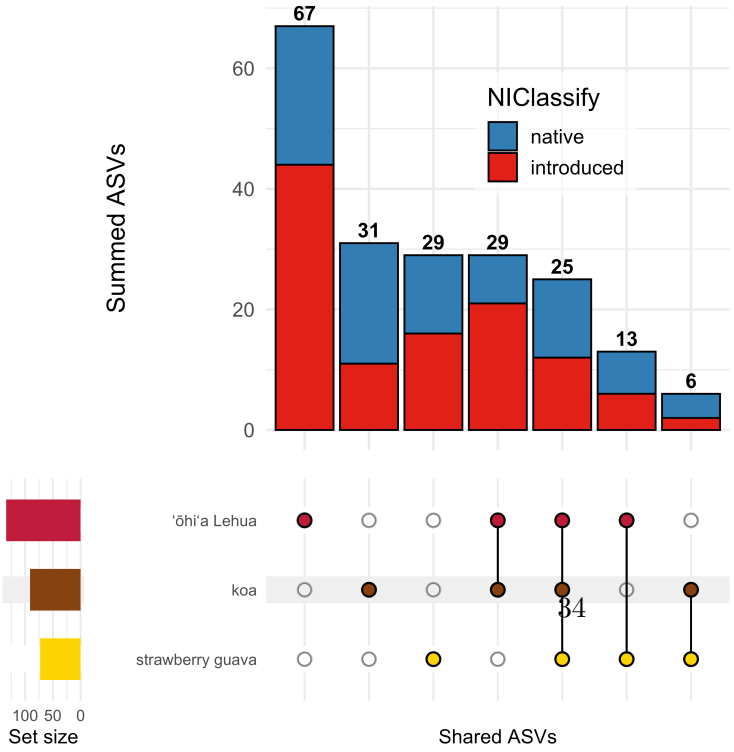
Data obtained through Ibis geolocation API, NIClassify predictions, BLAST mapping and species lookup

11 Comparing Invasive Status Predictions vs Blast or Ibis

12 Leke Transect Tree Comparison – Cleaned Code Chunk

13 Plot 3: UpSet plot by BLAST status as example





UpSet plots showing shared and unique zOTUs across co-occurring native (ōhi a, koa) and introduced (strawberry guava) trees on O ahu. Bars are colored by invasion status as inferred from BLAST matches (top) and NIClassify predictions (bottom).

```
    Asymptotic General Symmetry Test

data:  status by
     classifier (NiClassify, Blast)
     stratified by zotu
chi-squared = 51.969, df = 1, p-value = 5.638e-13
```

We found a significant difference in invasion status assignments between BLAST and NIClassify across zOTUs (asymptotic symmetry test, $^{2} = 51.97$, df $= 1$, p $= 5.6 \times 10^{13}$), demonstrating that the choice of classifier impacts ecological interpretation of community composition.

14 Supl. Plot : Heatmap of Confusion Matrix

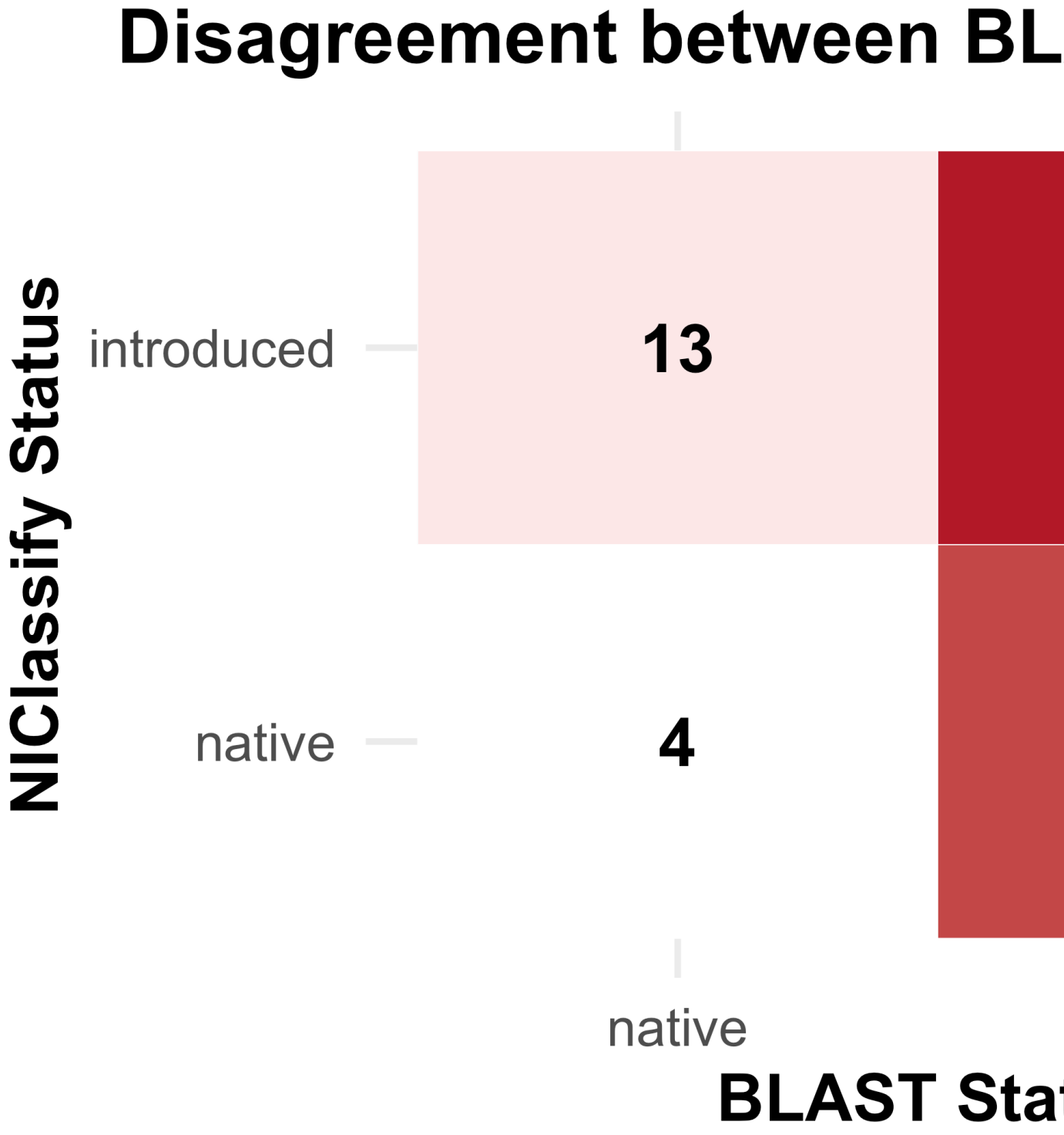
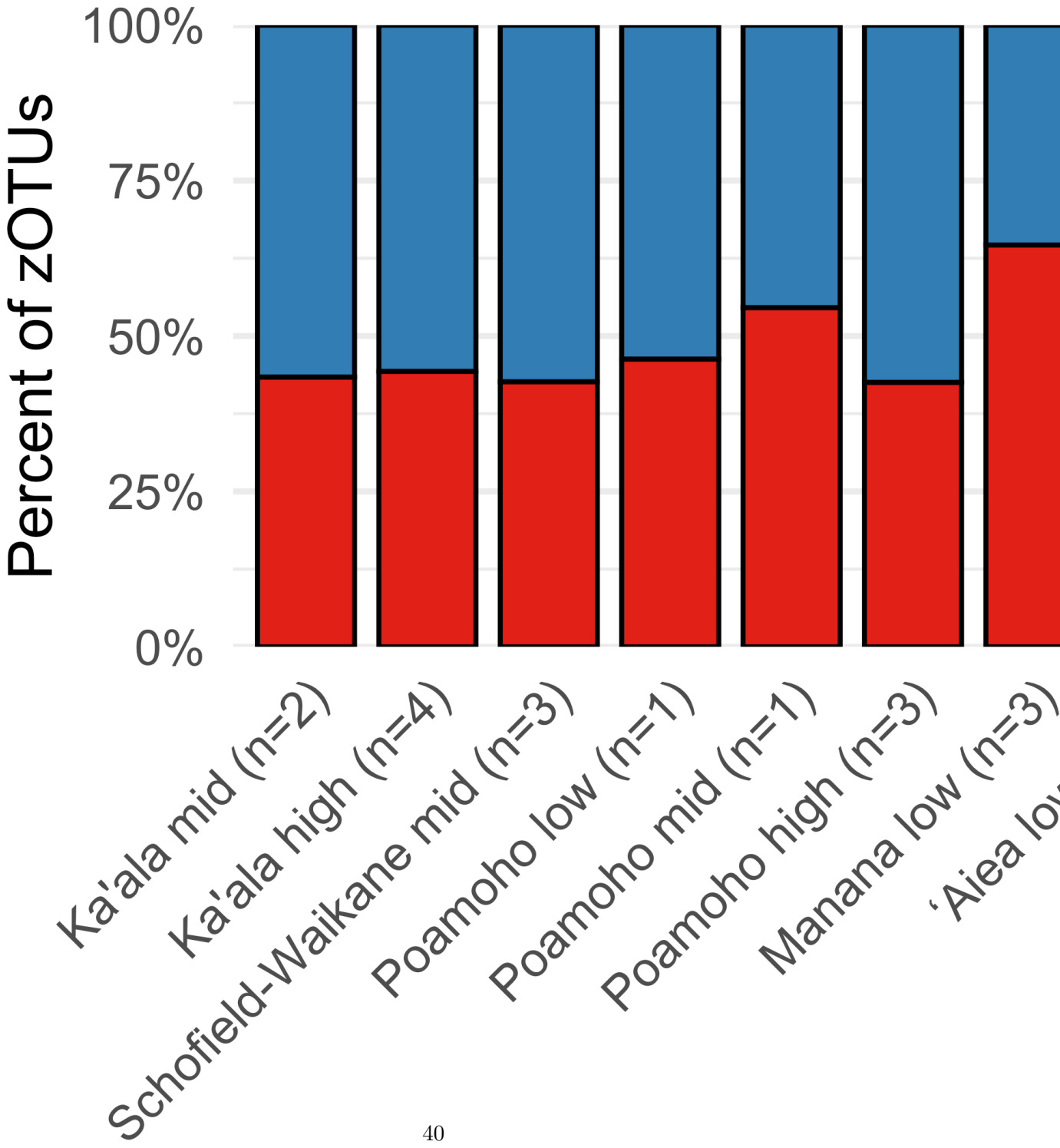


Figure X. Confusion matrix comparing invasion status assignments for zOTUs by BLAST (x-axis) and NIClassify (y-axis) on the Lekes transect. Cell values represent the number of zOTUs assigned to each combination. The majority of assignments are "introduced" by both classifiers, but a considerable number are classified as "native" by NIClassify and "introduced" by BLAST, indicating a significant difference in classifier behavior (symmetry test, $p < 0.001$).

15 Elevation Bands and Invasion Status across all Oahu ridges

16 Plot 4: Invasion among elevation bands



Stacked bar chart showing the percentage of unique zOTUs, classified as native (blue) or introduced (red), within each elevation band—low, mid, and high—across the five O ahu ridges: Ka‘ala, Schofield-Waikane, Poamoho, Manana, and Aiea. Elevation bands were defined by the 33rd and 67th percentile breaks of sampled elevations for each ridge, with the number of samples per band shown in parentheses on the x-axis (n). The consistently higher proportion of introduced zOTUs at lower elevations—particularly on Manana and Aiea—contrasts with a greater native zOTU representation at higher bands, suggesting an elevation-dependent invasion gradient in O ahu’s ōhi a forests.

```
Call:
geepack::geeglm(formula = cbind(introduced, native) ~ elev_band,
    family = binomial, data = inv_by_sample, id = ridge_id, corstr = "exchangeable")

 Coefficients:
             Estimate Std.err  Wald Pr(>|W|)
(Intercept)     0.3644  0.1635 4.968  0.02582 *
elev_bandmid   -0.5532  0.2460 5.057  0.02453 *
elev_bandhigh  -0.4602  0.1459 9.947  0.00161 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable
Estimated Scale Parameters:

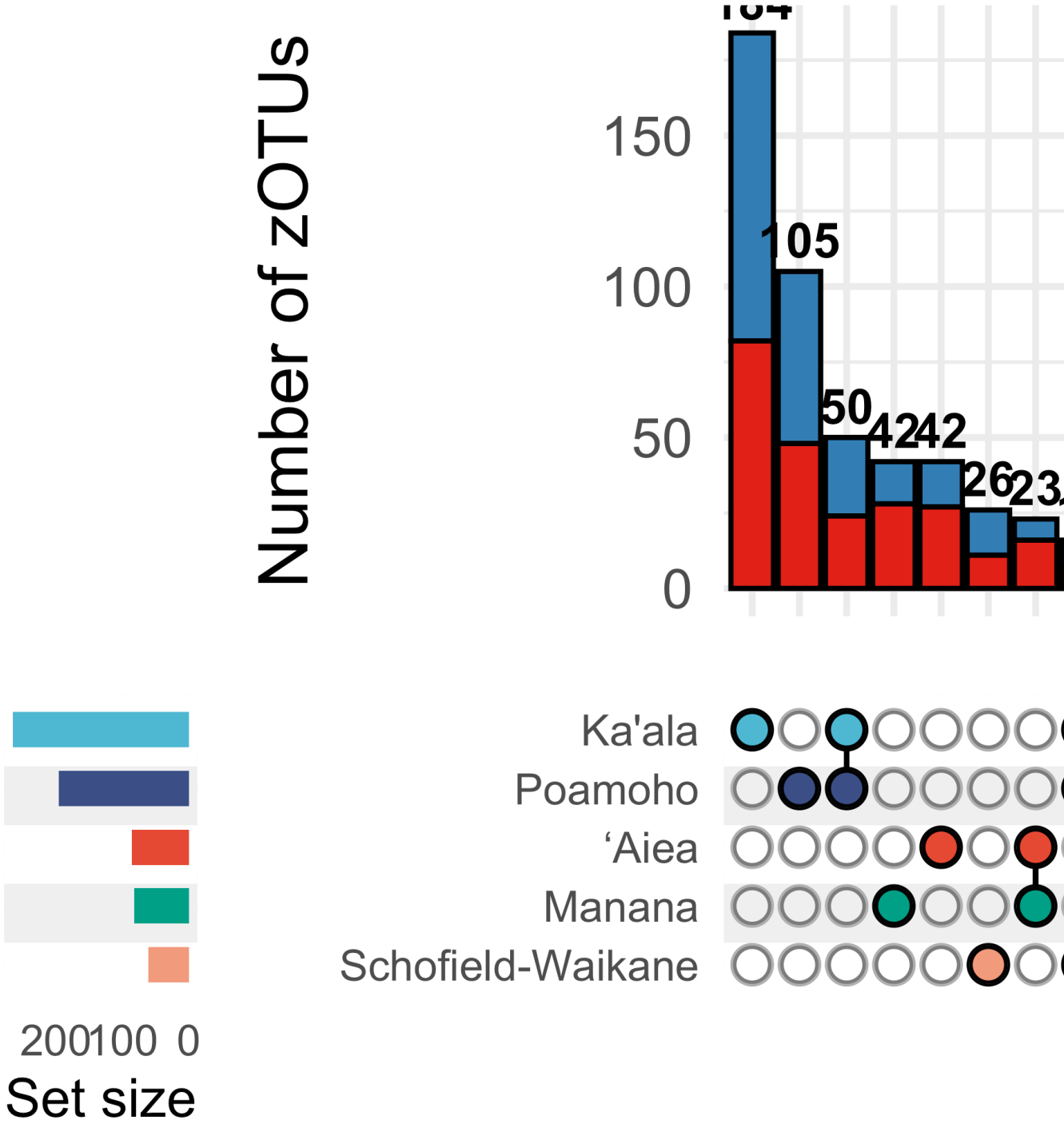
            Estimate  Std.err
(Intercept)  0.03849 0.003756
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha  -0.1953  0.1439
Number of clusters:   5  Maximum cluster size: 6
```

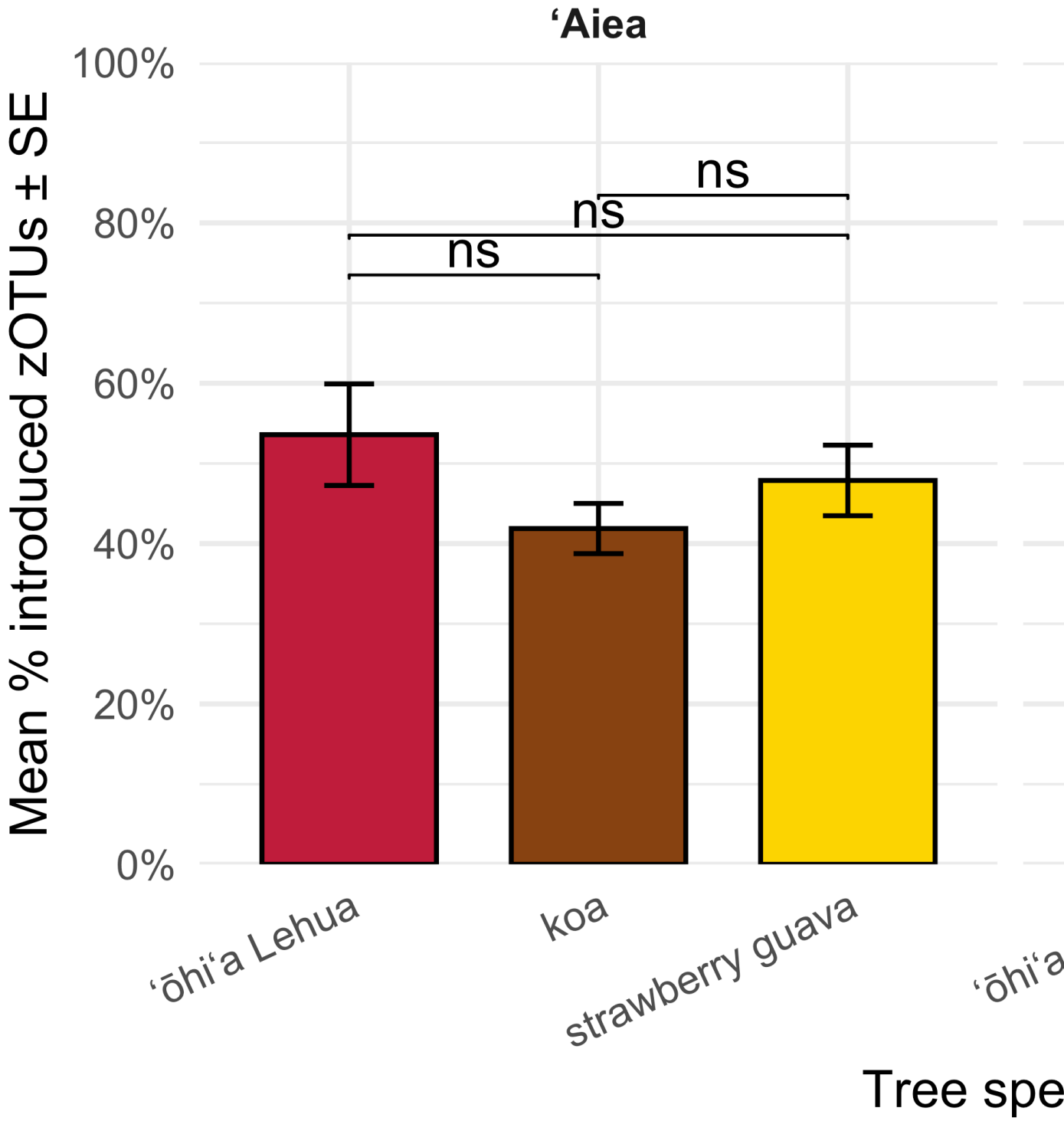
We used a binomial Generalized Estimating Equation (GEE) to test the effect of elevation band (low, mid, high) on the proportion of introduced zOTUs, with ridge as the clustering variable.

The model revealed a significant effect of elevation band (Wald $^2 = 9.95$, df = 2, p = 0.007), with both the mid (estimate = –0.55, p = 0.025) and high (estimate = –0.46, p = 0.0016) elevation bands showing significantly lower odds of introduced zOTUs compared to the low band.

17 Plot 5: UpSet plot of all five ridges



18 Plot 6: Statistical differences of invasion between tree species on Aiea and Manana ridges



19 Summary invasion status analysis

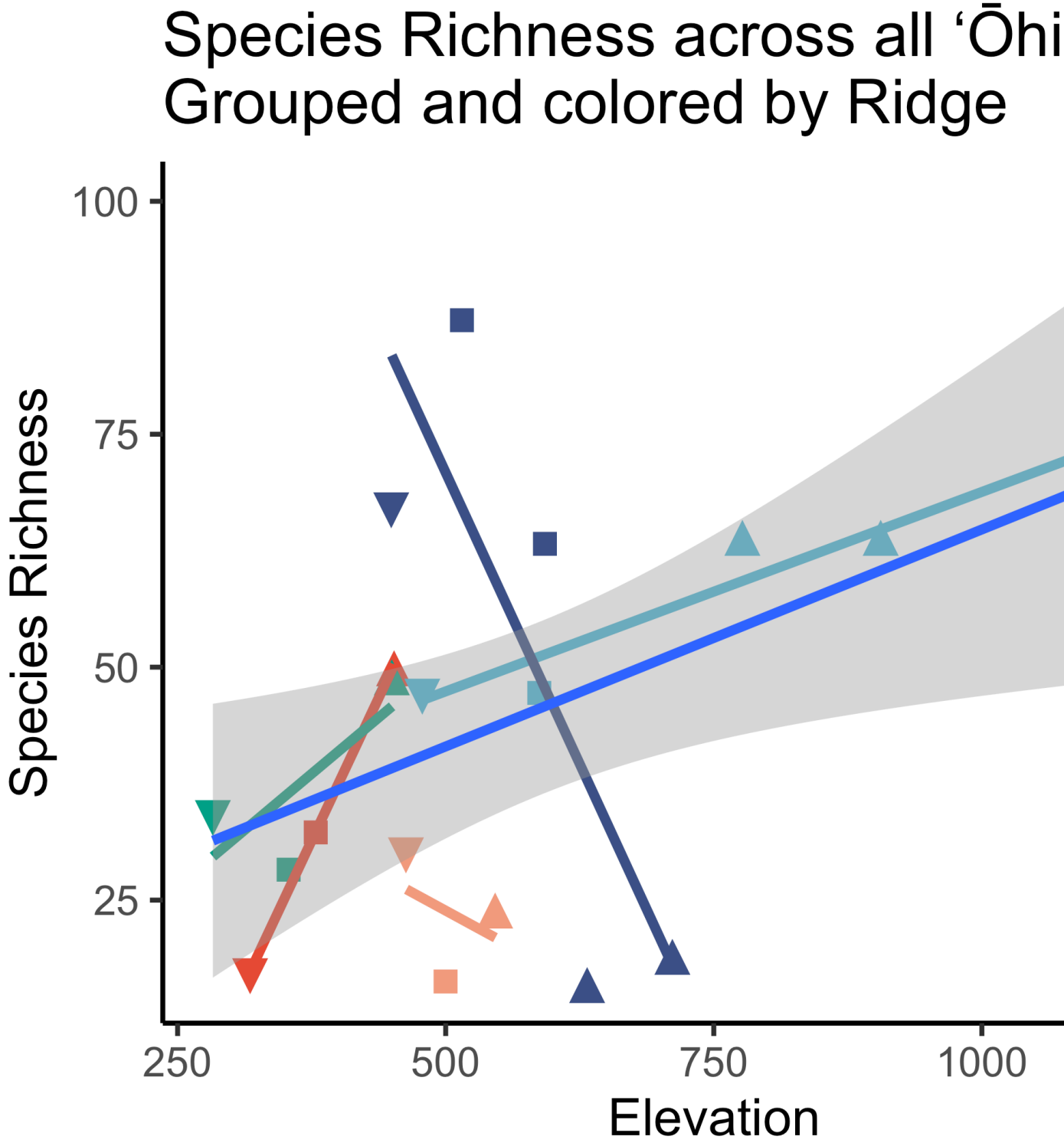
Data & status integration. We merged NIClassify predictions with curated BLAST/IBIS assignments, normalized case, and treated blanks as NA. Status was summarized per method (native / introduced / unknown) with a gt table. Tree comparisons (Leke transect). Presence/absence per ZOTU across ōhi a, koa, and guava; intersections visualized with ComplexUpset. Top bars colored by NIClassify or BLAST status. Classifier disagreement. For ZOTUs with both assignments, we tested symmetry of the 2×2 status table (native vs introduced) across classifiers using coin::symmetry_test with ZOTU as a blocking factor. Elevation effects (ōhi a on O ahu). We assigned elevation bands (terciles), computed % introduced per site×band, plotted stacked bars, and fit a binomial GEE (introduced vs native) with ridge as cluster. Ridge overlap. UpSet across five ridges with NIClassify coloring. Tree contrasts within ridges. Mean % introduced per tree, SE bars, pairwise tests with Holm correction (stars plotted).

Results support H2: introduced taxa are more prevalent at lower elevations, with a consistent decline up the gradient.

Combined with your TITAN (~500 m community breakpoint), this suggests a transition zone where lower-elevation assemblages (richer in introduced taxa) give way to higher-elevation assemblages.

H3 (site effects): UpSet/ridge panels and GEE clustering by ridge indicate ridge/site structure matters beyond elevation alone; we're not just seeing a smooth cline.

20 Plot 7: Species Richness across all ʻŌhiʻa samples from Oʻahu, grouped by elevation and ridge



```
Call:
lm(formula = richness ~ elevation_m, data = sr_hight)

Residuals:
   Min     1Q Median     3Q    Max
-32.34 -11.34   2.52   9.66  45.80

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  18.2022    11.3138    1.61    0.125
elevation_m   0.0466     0.0177    2.64    0.017 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.8 on 18 degrees of freedom
Multiple R-squared:  0.279, Adjusted R-squared:  0.238
F-statistic: 6.95 on 1 and 18 DF,  p-value: 0.0168


Call:
lm(formula = richness ~ elevation_m + sample_site, data = sr_hight)

Residuals:
   Min     1Q Median     3Q    Max
-36.03  -7.01  -0.04   9.86  38.75

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)
(Intercept)                   24.1931    15.5318    1.56     0.14
elevation_m                    0.0239     0.0274    0.87     0.40
sample_siteKa'ala             18.0281    19.1087    0.94     0.36
sample_siteManana              4.4924    16.1914    0.28     0.79
sample_sitePoamoho            12.7554    15.4526    0.83     0.42
sample_siteSchofield-Waikane -12.5387    16.5147   -0.76     0.46

Residual standard error: 19.8 on 14 degrees of freedom
Multiple R-squared:  0.438, Adjusted R-squared:  0.237
F-statistic: 2.18 on 5 and 14 DF,  p-value: 0.115


Linear mixed model fit by REML ['lmerMod']
Formula: richness ~ elevation_m + (1 | sample_site)
   Data: sr_hight
```

```
REML criterion at convergence: 175.6

Scaled residuals:
   Min     1Q Median     3Q    Max
-1.634 -0.573  0.127  0.488  2.313

Random effects:
 Groups      Name        Variance Std.Dev.
 sample_site (Intercept)   0       0.0
 Residual                 392      19.8
Number of obs: 20, groups:  sample_site, 5

Fixed effects:
            Estimate Std. Error t value
(Intercept)  18.2022    11.3138    1.61
elevation_m   0.0466     0.0177    2.64

Correlation of Fixed Effects:
            (Intr)
elevation_m -0.920
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

Species richness increased significantly with elevation across all O ahu ōhi a samples (linear regression: $= 0.047$ species/m, $R^2 = 0.28$, $p = 0.017$). Including ridge as an additional factor in multiple regression and mixed-effects models did not improve model fit, and the elevation effect was robust to site effects (see Tables).

```
N samples: 34  | N taxa (pre-filter): 854


   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    283     360     452     524     577    1218


N taxa kept (occ >= 3): 115


         ienv.cp zenv.cp freq maxgrp IndVal zscore    5%   10%   50%   90%
zotu1016   457.7   457.7    4      1  21.05   1.56 335.7 378.9 452.0 465.1
zotu1017   457.7   457.7    6      1  31.58   2.57 366.4 449.9 457.7 465.1
zotu1045   452.0   452.0    3      1  17.65   1.82 317.6 317.6 378.9 457.7
zotu1052   452.0   452.0    6      1  33.33   3.08 353.9 366.4 452.0 465.1
zotu1054   612.3   612.3    4      2  41.57   5.20 414.2 449.7 612.3 652.1
```

zotu1066	414.2	414.2	3	2	13.64	0.65	413.6	414.2	449.7	514.1
zotu109	590.2	590.2	3	2	37.50	5.60	530.5	554.0	590.2	994.6
zotu1125	457.7	457.7	3	2	20.00	2.07	456.6	457.7	457.7	744.2
zotu1167	449.9	449.9	3	1	21.43	2.37	353.6	353.9	449.9	451.0
zotu118	566.9	507.5	7	2	54.28	4.60	499.3	507.5	566.9	744.2
zotu12	449.9	449.9	5	1	33.33	2.81	317.6	317.6	449.9	451.0
zotu122	449.9	457.7	6	1	26.03	1.36	353.9	353.9	451.0	457.7
zotu123	744.2	507.5	5	2	39.34	3.05	414.2	496.7	566.9	744.2
zotu124	470.8	457.7	5	2	35.71	3.61	465.1	465.1	569.5	841.4
zotu1265	449.9	449.9	4	1	28.57	3.31	299.4	315.9	378.9	451.0
zotu1270	744.2	566.9	3	2	39.02	3.69	530.7	543.8	678.3	994.6
zotu129	481.0	481.0	3	2	23.08	2.56	353.9	467.9	489.2	994.6
zotu1299	457.7	470.8	3	1	15.79	1.12	365.7	414.4	451.0	457.7
zotu13	470.8	470.8	4	2	16.38	0.88	353.9	353.9	470.8	930.1
zotu144	744.2	590.2	3	2	38.96	4.18	515.1	554.0	590.2	744.2
zotu1451	449.9	457.7	5	1	27.48	2.33	353.9	366.4	449.9	457.7
zotu1452	457.7	457.7	5	2	33.33	2.85	457.7	457.7	457.7	590.2
zotu1454	457.7	457.7	6	1	31.58	3.09	353.9	366.4	457.7	465.1
zotu147	530.7	566.9	5	1	20.83	0.89	335.7	353.9	457.7	530.7
zotu1481	491.8	491.8	8	2	56.63	5.34	457.7	457.7	491.8	714.7
zotu1494	449.7	449.9	3	2	14.29	0.64	353.0	414.4	449.9	451.5
zotu15	470.8	470.8	5	2	28.57	1.82	414.2	414.2	470.8	523.0
zotu151	452.0	452.0	7	1	38.89	2.45	317.6	335.7	457.7	465.1
zotu1541	414.2	414.2	3	1	25.00	3.03	299.4	300.2	378.9	414.2
zotu1543	566.9	566.9	3	2	22.06	2.69	464.1	465.1	551.4	704.4
zotu1548	449.9	449.9	5	1	35.71	3.69	300.2	317.6	378.9	451.0
zotu1557	452.0	452.0	8	1	37.34	2.69	300.2	317.6	452.0	530.7
zotu1564	457.7	457.7	4	1	21.05	1.93	353.9	353.9	451.0	457.7
zotu159	744.2	507.5	3	2	18.57	0.89	366.4	378.9	507.5	841.4
zotu16	470.8	470.8	3	2	21.43	2.49	464.1	465.1	515.0	936.5
zotu1611	457.7	457.7	7	1	36.84	3.21	366.4	414.2	457.7	465.1
zotu168	612.3	491.8	5	2	41.99	3.99	481.6	489.1	491.8	612.3
zotu1687	507.5	507.5	3	2	27.27	3.92	489.2	496.7	507.5	671.6
zotu17	457.7	452.0	12	2	64.77	5.11	457.7	457.7	481.0	841.4
zotu18	366.4	366.4	6	1	30.06	1.67	353.9	353.9	366.4	530.7
zotu184	744.2	566.9	3	2	39.85	3.61	530.7	543.8	684.9	994.6
zotu1848	457.7	457.7	3	1	15.79	1.29	334.8	353.9	451.0	457.7
zotu19	457.7	457.7	4	2	26.67	3.45	457.7	457.7	457.7	905.9
zotu1903	449.9	449.9	5	1	35.71	3.60	353.9	366.4	449.9	451.0
zotu195	353.9	457.7	4	1	21.70	1.28	353.9	353.9	449.9	457.7
zotu1963	378.9	378.9	3	1	27.27	3.42	317.6	317.6	372.6	414.4
zotu1964	414.2	414.2	4	1	33.33	3.56	353.9	353.9	414.2	414.4
zotu198	449.9	449.9	4	1	28.57	2.80	282.9	300.2	449.7	451.0

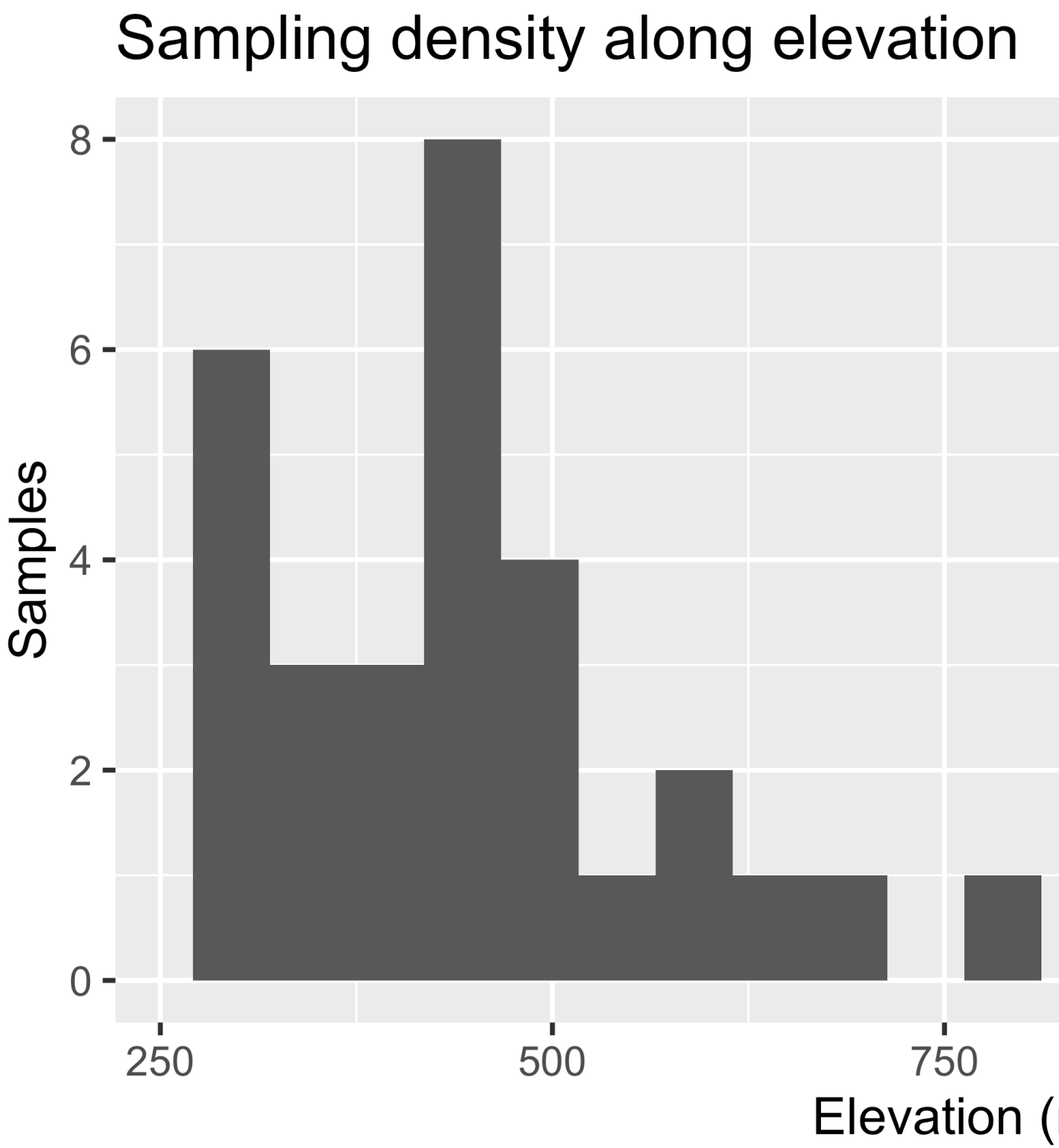
zotu20	335.7	449.7	23	1	97.21	4.08	317.6	318.4	449.9	465.1
zotu201	452.0	452.0	7	2	35.24	2.20	450.8	451.0	456.6	930.1
zotu203	671.6	671.6	6	2	56.63	5.40	465.1	470.8	650.8	905.9
zotu2114	414.2	414.2	3	1	25.00	3.25	353.9	353.9	378.9	414.4
zotu213	744.2	744.2	3	2	38.66	4.24	483.4	499.4	671.6	905.9
zotu2146	378.9	378.9	4	1	36.36	4.59	335.7	353.9	378.9	414.4
zotu22	507.5	470.8	7	2	45.34	2.94	414.2	414.2	499.4	547.5
zotu223	470.8	470.8	7	2	35.51	2.61	414.2	414.2	467.9	530.7
zotu2238	457.7	457.7	3	1	15.79	1.35	366.4	378.9	449.7	457.7
zotu229	590.2	590.2	3	2	24.96	2.57	456.6	457.7	590.2	841.4
zotu23	353.9	353.9	4	1	57.14	7.90	300.2	300.2	353.9	366.4
zotu240	744.2	671.6	4	2	57.91	4.63	522.6	523.0	744.2	841.4
zotu241	449.9	449.9	4	1	28.57	2.85	353.9	366.4	449.9	451.0
zotu249	590.2	590.2	4	2	35.11	3.82	414.2	414.2	590.2	744.2
zotu25	671.6	507.5	13	2	69.57	4.07	457.7	457.7	556.6	744.2
zotu29	470.8	457.7	11	2	70.00	5.53	465.1	465.1	507.5	714.7
zotu294	451.0	451.0	7	1	33.88	2.82	335.7	353.9	451.0	489.1
zotu305	507.5	507.5	3	2	27.27	4.09	478.3	496.7	507.5	776.9
zotu310	507.5	507.5	3	2	27.27	3.48	353.9	496.7	507.5	631.9
zotu313	744.2	671.6	4	2	58.90	5.81	535.7	551.4	704.4	994.6
zotu317	470.8	470.8	3	1	15.00	0.84	353.9	353.9	457.7	473.5
zotu32	457.7	457.7	5	1	20.98	1.11	353.9	353.9	449.9	530.7
zotu33	378.9	378.9	3	1	30.00	3.88	300.2	317.6	378.9	414.4
zotu331	449.9	449.9	3	1	21.43	2.35	353.9	366.4	449.9	451.0
zotu34	317.6	317.6	3	1	38.33	4.48	317.6	317.6	335.7	457.7
zotu354	744.2	507.5	4	2	39.96	3.90	496.7	499.4	590.2	744.2
zotu384	414.2	414.2	3	2	13.64	0.64	401.7	414.2	465.1	489.1
zotu388	378.9	378.9	3	2	13.04	0.51	353.9	366.4	449.9	530.7
zotu406	414.2	414.2	8	1	53.49	4.19	317.6	317.6	414.2	457.7
zotu42	566.9	507.5	8	2	52.99	3.91	457.7	496.7	566.9	744.2
zotu420	449.9	449.9	3	2	15.79	1.12	414.4	415.3	451.0	507.5
zotu427	566.9	566.9	3	2	33.33	5.00	535.7	543.8	566.9	653.1
zotu438	335.7	452.0	10	1	68.15	4.64	335.7	335.7	348.2	452.0
zotu46	744.2	491.8	6	2	54.53	4.25	481.6	489.1	578.6	744.2
zotu482	481.0	481.0	7	2	53.85	4.71	467.9	473.5	481.0	744.2
zotu484	366.4	366.4	7	1	62.47	5.58	300.2	300.2	366.4	366.4
zotu485	744.2	744.2	6	2	55.21	4.65	456.2	457.7	631.9	768.9
zotu49	366.4	452.0	6	2	24.00	0.84	366.4	366.4	452.0	590.2
zotu529	335.7	457.7	7	1	42.31	2.50	300.2	300.2	353.9	457.7
zotu53	414.2	414.2	5	1	41.67	4.15	353.9	414.2	414.2	414.4
zotu56	449.7	414.2	3	2	14.29	0.49	335.7	414.4	450.3	515.0
zotu563	449.9	449.9	7	1	39.03	3.43	317.6	335.7	449.9	457.7
zotu578	457.7	457.7	5	2	33.33	2.89	457.7	457.7	457.7	744.2

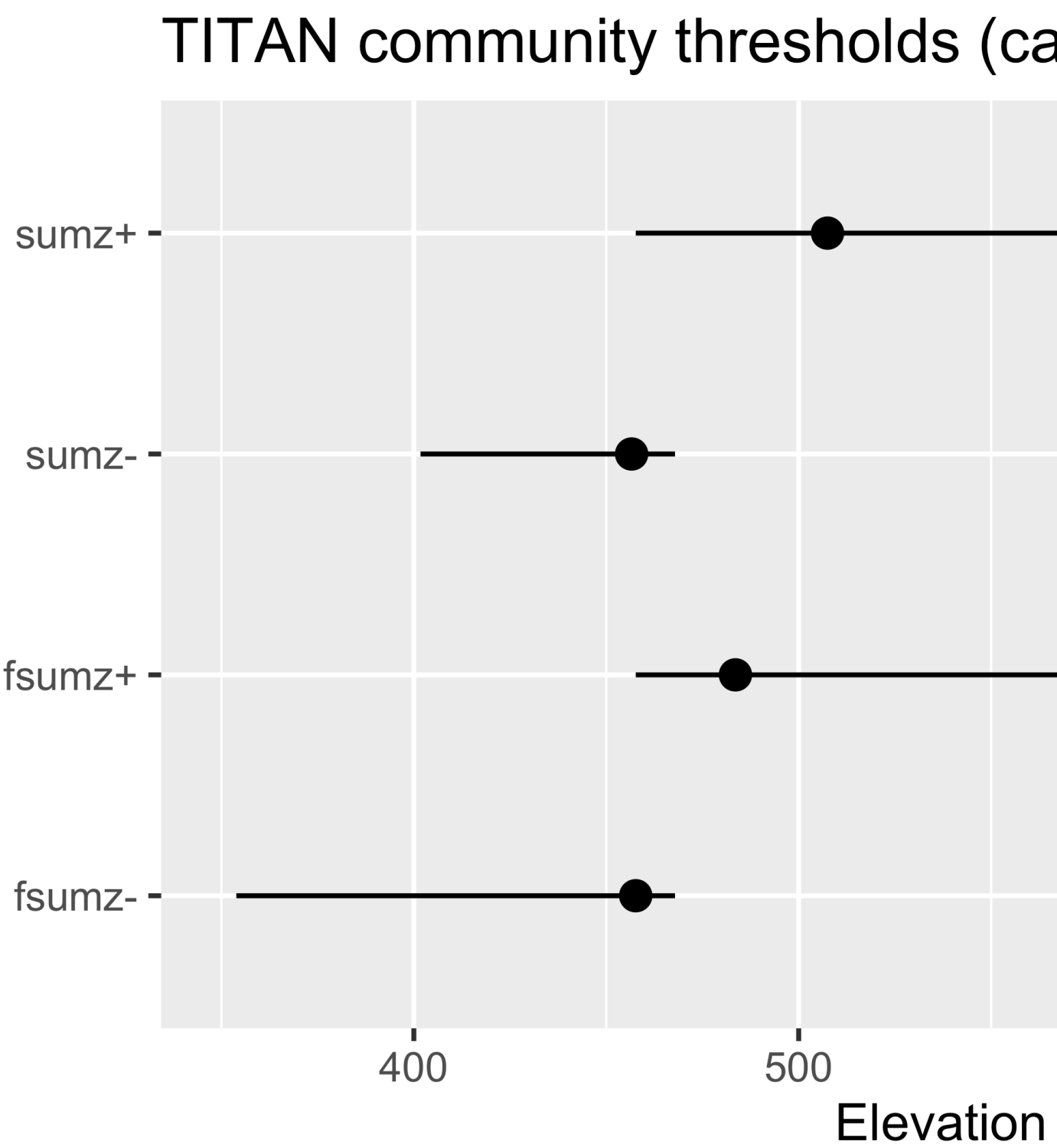
```
zotu582    378.9   378.9    4      1  40.00   4.78 300.2 317.6 372.6 414.2
zotu60     566.9   566.9    5      2  41.86   4.64 465.1 470.8 566.9 744.2
zotu635    414.2   414.2    4      2  18.18   0.98 414.2 414.2 481.0 994.6
zotu664    530.7   566.9    6      1  25.00   1.31 353.9 353.9 452.0 530.7
zotu67     671.6   457.7   12      2  73.21   4.97 457.7 457.7 507.5 744.2
zotu681    671.6   671.6    3      2  29.17   2.86 414.4 449.7 652.1 905.9
zotu69     507.5   507.5    9      2  58.32   5.03 465.1 470.8 507.5 662.6
zotu690    507.5   507.5    3      2  16.44   1.21 353.9 414.2 507.5 841.4
zotu7      457.7   452.0   17      1  41.30   0.71 317.6 317.6 414.2 930.1
zotu70     378.9   378.9   11      1  69.58   4.68 317.6 347.0 414.2 457.7
zotu71     452.0   452.0   13      2  40.87   2.05 317.6 335.7 451.0 671.6
zotu717    566.9   449.9    6      1  24.00   1.15 317.6 347.0 449.9 566.9
zotu72     566.9   566.9    6      2  51.87   5.40 465.1 470.8 566.9 783.4
zotu74     507.5   507.5    6      2  44.24   4.70 465.1 465.1 507.5 808.6
zotu746    452.0   452.0    3      1  17.65   1.72 334.8 353.9 366.4 457.7
zotu754    491.8   491.8    5      2  41.67   4.69 481.6 481.6 491.8 566.9
zotu770    481.0   481.0    4      2  22.75   1.78 317.6 335.7 481.0 566.9
zotu797    452.0   452.0    3      1  17.65   1.73 353.9 353.9 452.0 458.4
zotu80     744.2   457.7    7      2  46.99   2.74 414.2 414.2 537.2 749.4
zotu87     744.2   507.5    5      2  54.16   4.71 499.4 507.5 631.9 841.4
zotu9      353.9   452.0   14      1  73.93   4.16 335.7 335.7 457.7 465.1
zotu900    414.2   414.2    3      1  25.00   3.00 317.6 353.9 378.9 414.4
zotu955    317.6   317.6    4      1  32.56   3.12 300.2 300.2 353.9 451.0
zotu957    744.2   590.2    3      2  39.94   4.10 530.5 554.0 609.8 841.4
           95% purity reliability z.median filter
zotu1016  465.1  0.916       0.354    2.037      0
zotu1017  467.9  0.928       0.642    2.703      0
zotu1045  465.1  0.906       0.350    1.949      0
zotu1052  467.9  0.992       0.748    3.138      0
zotu1054  652.1  0.932       0.616    3.965      0
zotu1066  530.7  0.684       0.146    1.295      0
zotu109  1083.3  0.970       0.730    5.276      0
zotu1125  841.4  0.950       0.442    2.514      0
zotu1167  451.0  0.940       0.418    2.398      0
zotu118   768.9  1.000       0.954    5.353      2
zotu12    451.0  0.992       0.726    3.023      0
zotu122   465.2  0.680       0.438    1.834      0
zotu123   905.9  0.984       0.770    3.755      0
zotu124   841.4  0.994       0.888    4.408      0
zotu1265  451.0  0.980       0.706    3.066      0
zotu1270 1083.3  0.968       0.730    4.910      0
zotu129  1083.3  0.936       0.578    3.010      0
zotu1299  464.1  0.634       0.238    1.580      0
```

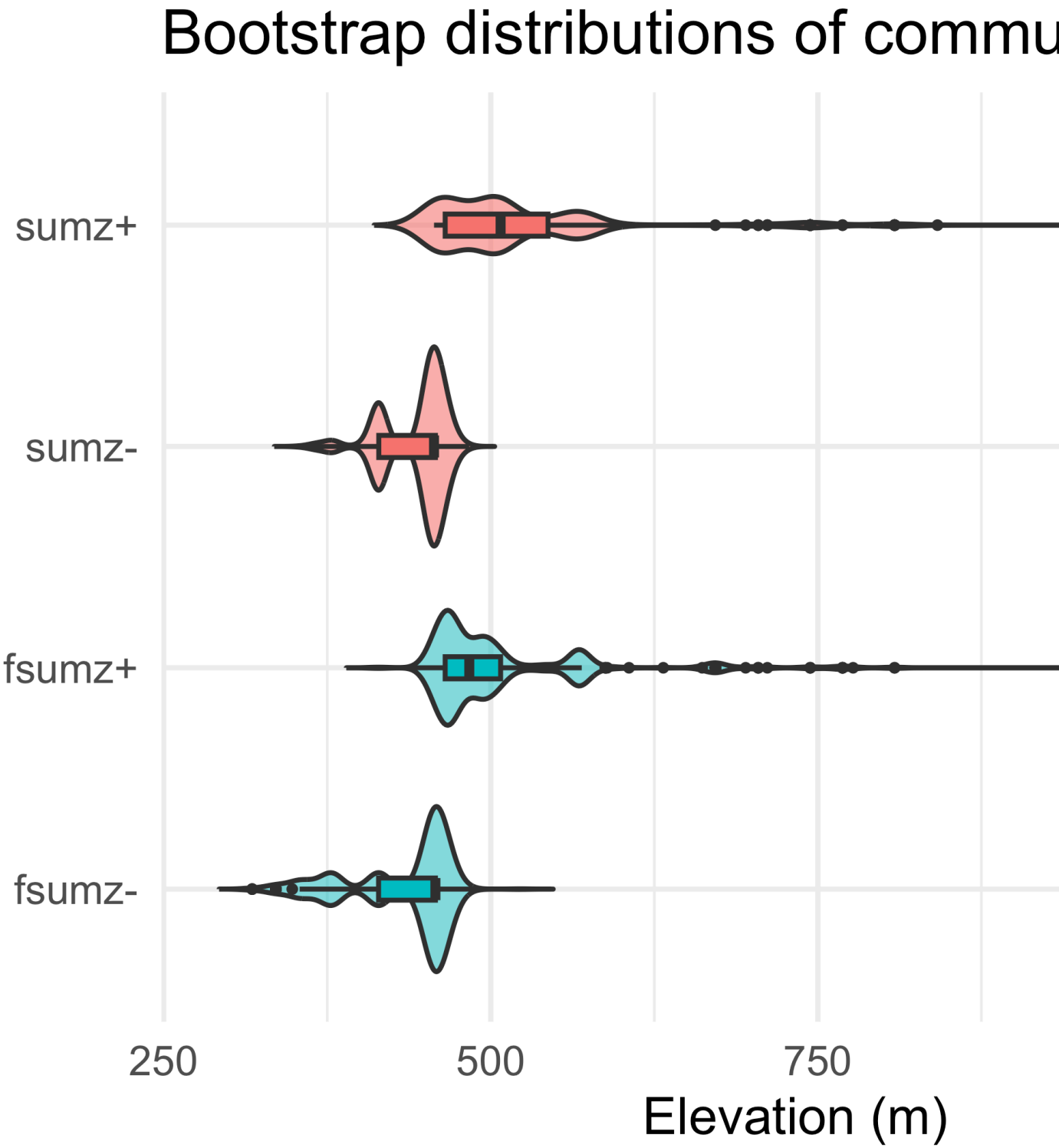
zotu13	1083.3	0.750	0.332	1.786	0
zotu144	770.8	0.956	0.736	4.970	0
zotu1451	465.1	0.960	0.666	2.901	0
zotu1452	808.6	0.990	0.802	3.385	0
zotu1454	467.9	0.936	0.662	2.753	0
zotu147	551.4	0.574	0.230	1.548	0
zotu1481	841.4	1.000	0.980	5.292	2
zotu1494	590.2	0.596	0.150	1.314	0
zotu15	633.8	0.950	0.560	2.468	0
zotu151	467.9	0.994	0.742	2.646	0
zotu1541	414.4	0.958	0.676	3.708	0
zotu1543	744.2	0.954	0.488	2.787	0
zotu1548	451.0	0.990	0.814	3.814	0
zotu1557	551.4	0.988	0.714	2.854	0
zotu1564	465.1	0.932	0.442	2.233	0
zotu159	841.4	0.814	0.384	1.983	0
zotu16	1083.3	0.954	0.492	2.843	0
zotu1611	467.9	0.978	0.794	3.176	0
zotu168	652.1	0.998	0.932	5.011	0
zotu1687	905.9	0.964	0.590	3.348	0
zotu17	841.4	1.000	1.000	5.449	2
zotu18	551.4	0.930	0.430	1.997	0
zotu184	1083.3	0.968	0.730	4.924	0
zotu1848	465.1	0.810	0.216	1.607	0
zotu19	1062.8	0.976	0.572	2.999	0
zotu1903	451.0	0.994	0.838	3.830	0
zotu195	465.1	0.934	0.456	2.265	0
zotu1963	414.4	0.942	0.582	3.505	0
zotu1964	414.4	0.992	0.762	3.976	0
zotu198	451.0	0.980	0.616	2.838	0
zotu20	467.9	1.000	0.998	4.261	1
zotu201	994.6	0.918	0.570	2.530	0
zotu203	1038.8	0.998	0.950	5.260	2
zotu2114	414.4	0.970	0.626	3.646	0
zotu213	1083.3	0.952	0.694	4.283	0
zotu2146	414.4	0.978	0.798	4.433	0
zotu22	590.2	0.992	0.764	3.241	0
zotu223	907.1	0.970	0.724	3.043	0
zotu2238	465.1	0.818	0.248	1.863	0
zotu229	841.4	0.952	0.526	2.760	0
zotu23	366.4	0.992	0.916	6.597	0
zotu240	1083.3	0.988	0.870	6.026	0
zotu241	451.0	0.974	0.666	3.006	0

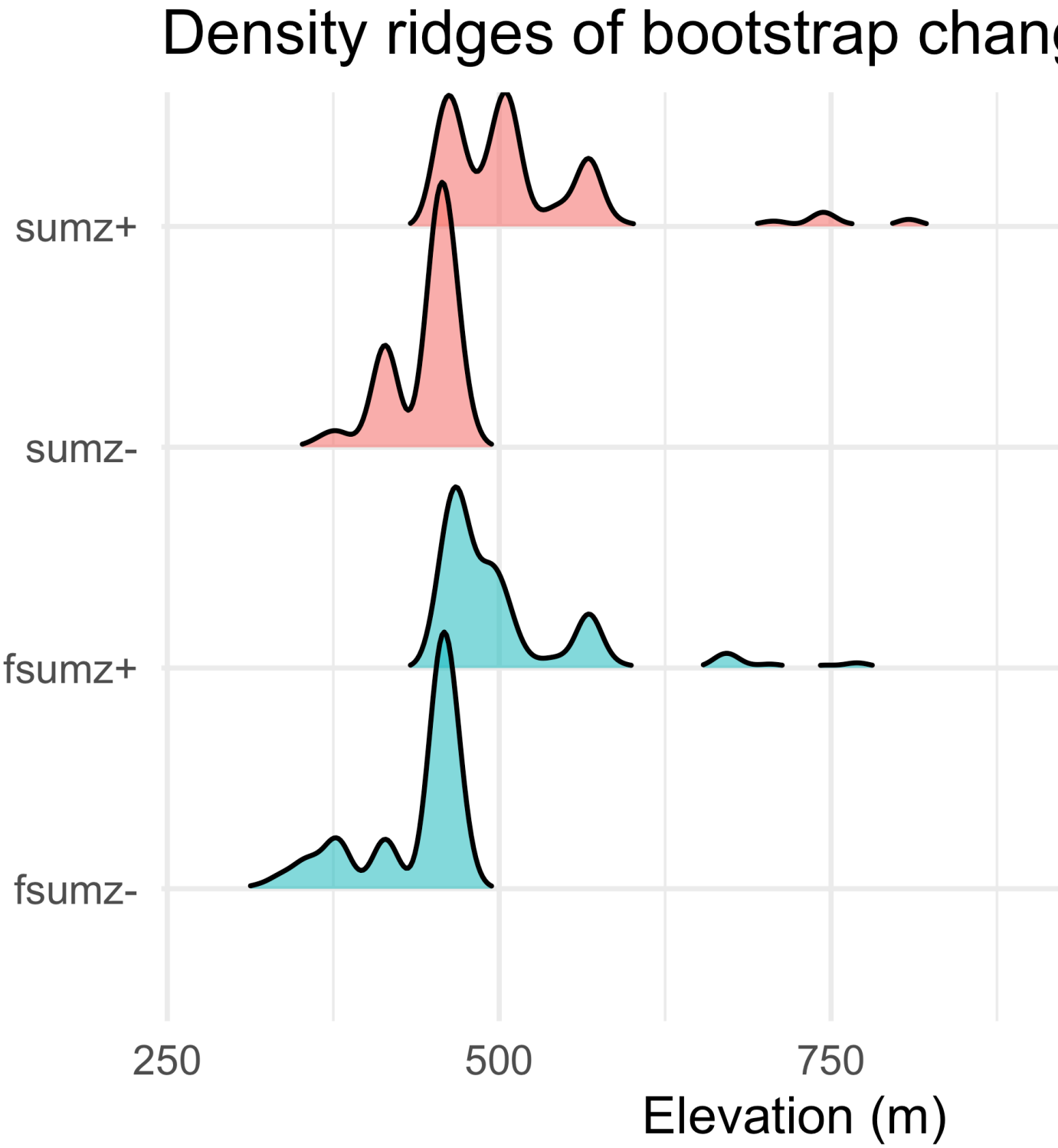
zotu249	744.2	0.940	0.664	3.755	0
zotu25	808.6	1.000	0.990	4.845	2
zotu29	768.9	1.000	0.998	6.114	2
zotu294	489.1	0.984	0.744	3.108	0
zotu305	841.4	0.950	0.626	3.775	0
zotu310	744.2	0.944	0.618	3.658	0
zotu313	1083.3	0.986	0.890	6.071	0
zotu317	481.6	0.778	0.138	1.305	0
zotu32	530.7	0.726	0.286	1.540	0
zotu33	414.4	0.952	0.642	3.417	0
zotu331	451.0	0.944	0.424	2.272	0
zotu34	457.7	0.908	0.542	3.099	0
zotu354	744.3	0.988	0.808	4.651	0
zotu384	671.6	0.716	0.150	1.317	0
zotu388	530.7	0.358	0.168	1.284	0
zotu406	465.1	1.000	0.948	4.554	0
zotu42	744.2	1.000	0.946	4.861	0
zotu420	507.5	0.676	0.184	1.321	0
zotu427	744.2	0.974	0.720	4.492	0
zotu438	457.7	1.000	0.986	5.510	1
zotu46	808.6	0.998	0.960	5.607	2
zotu482	841.4	0.998	0.984	5.020	2
zotu484	378.9	1.000	0.966	6.261	1
zotu485	808.6	0.986	0.814	3.998	0
zotu49	612.3	0.918	0.416	1.980	0
zotu529	465.1	0.992	0.894	3.977	0
zotu53	414.4	0.990	0.890	4.303	0
zotu56	530.7	0.700	0.114	1.204	0
zotu563	465.1	0.994	0.888	3.942	0
zotu578	768.9	0.992	0.808	3.374	0
zotu582	414.4	0.990	0.832	4.657	0
zotu60	905.9	0.996	0.888	4.658	0
zotu635	1083.3	0.902	0.364	1.881	0
zotu664	551.4	0.756	0.392	1.999	0
zotu67	768.9	1.000	0.998	6.021	2
zotu681	1038.8	0.892	0.436	2.474	0
zotu69	704.4	1.000	0.986	5.473	2
zotu690	841.4	0.854	0.330	1.847	0
zotu7	994.6	0.562	0.428	1.731	0
zotu70	465.1	1.000	1.000	4.997	1
zotu71	930.1	0.652	0.526	2.114	0
zotu717	569.5	0.860	0.414	1.775	0
zotu72	841.4	0.998	0.956	5.354	2

```
zotu74    841.4  0.998       0.940    4.868      0
zotu746   465.1  0.882       0.302    1.923      0
zotu754   612.3  0.998       0.882    4.350      0
zotu770   673.3  0.808       0.446    2.253      0
zotu797   465.1  0.912       0.260    1.840      0
zotu80    808.6  0.990       0.814    3.627      0
zotu87   1083.3  0.992       0.950    6.098      2
zotu9     467.9  1.000       1.000    5.915      1
zotu900   414.4  0.958       0.648    3.559      0
zotu955   451.0  0.980       0.646    3.648      0
zotu957   857.6  0.968       0.754    5.133      0
          cp  0.05  0.10  0.50  0.90  0.95
sumz-  457.7 401.7 414.2 456.6 465.1 467.9
sumz+  507.5 457.7 457.7 507.5 569.5 744.2
fsumz- 452.0 353.9 366.4 457.7 465.1 467.9
fsumz+ 491.8 457.7 457.7 483.6 569.5 671.6
```


Sampling density along elevation
Samples
8
6
4
2
0
250
500
750
Elevation (

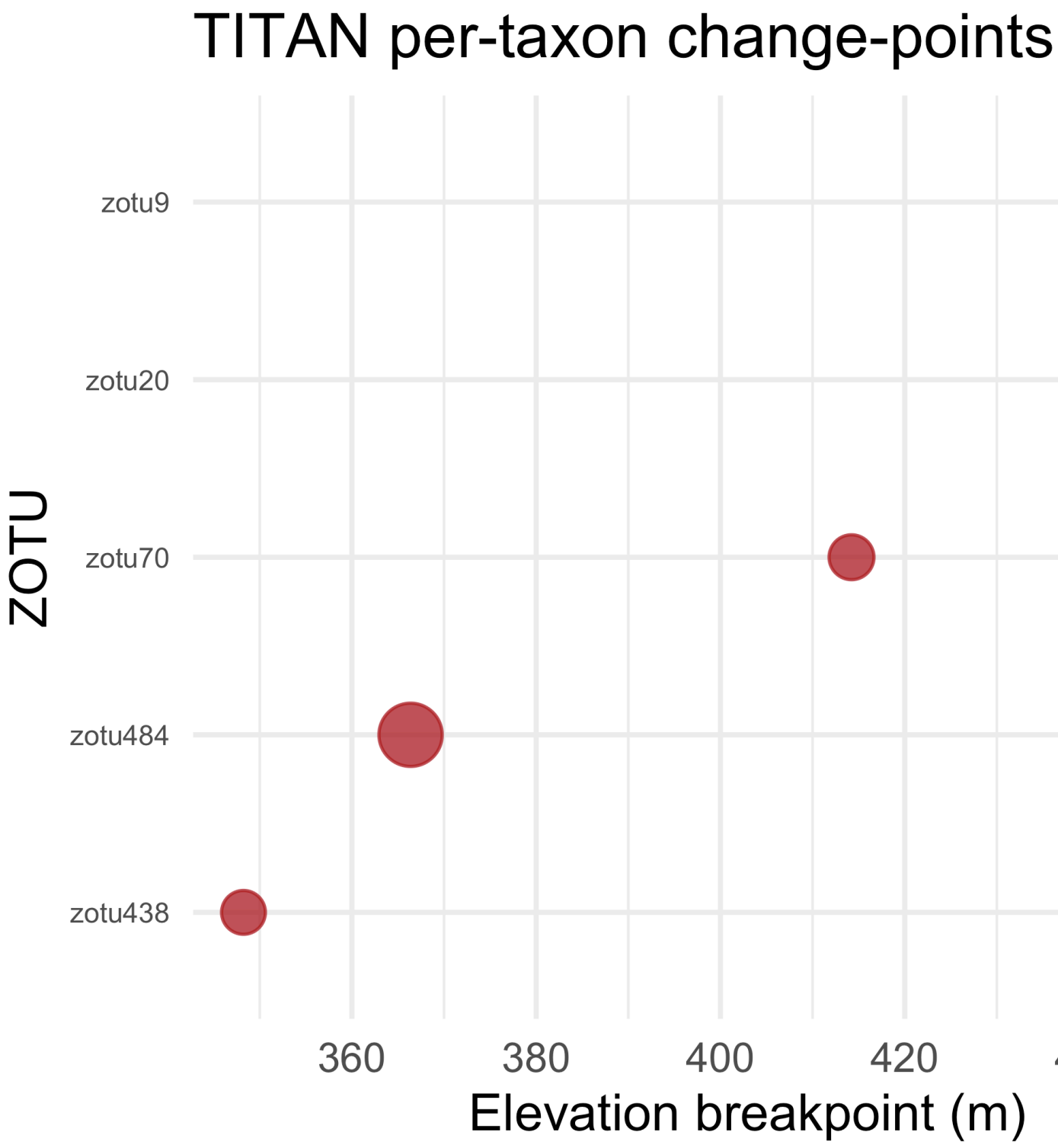

TITAN community thresholds (ca
sumz+
sumz-
fsumz+
fsumz-
400
500
Elevation


Bootstrap distributions of commu
sumz+
sumz-
fsumz+
fsumz-
250
500
750
Elevation (m)


Density ridges of bootstrap chang
sumz+
sumz-
fsumz+
fsumz-
250
500
750
Elevation (m)


TITAN community thresholds (ca
sumz+
sumz-
400
500
Elevation

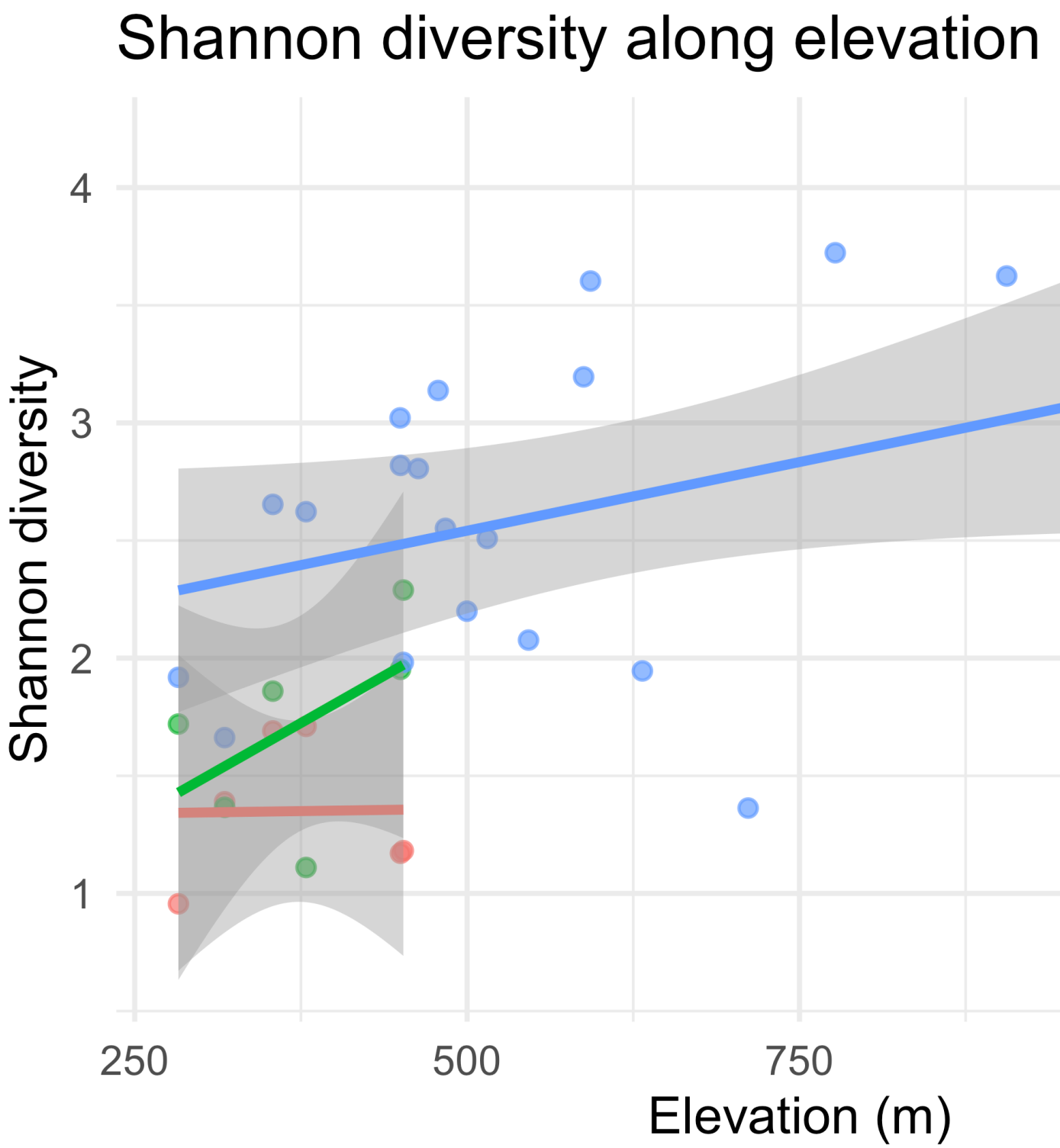
21 Plot 8: Titan per-taxon change-points

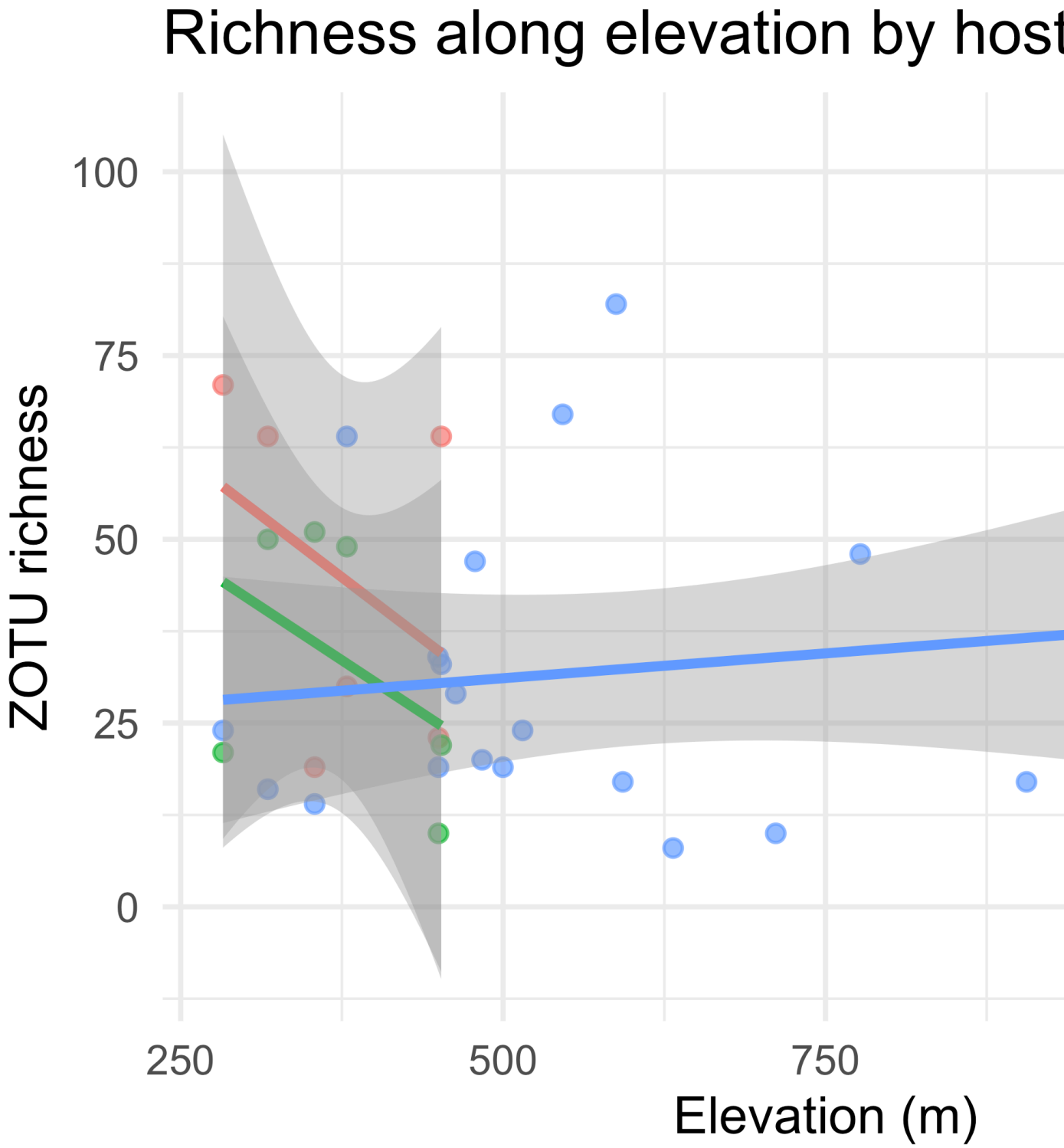


```
[1] "matrix" "array"
```

```
 num [1:34, 1:854] 723 0 0 0 0 0 0 0 0 0 ...
 - attr(*, "dimnames")=List of 2
  ..$ : chr [1:34] "KA4" "KT0" "POA3" "KA6" ...
  ..$ : chr [1:854] "zotu1000" "zotu1001" "zotu1003" "zotu1008" ...
```

```
  sample_id richness.x shannon.x sample_site elevation elevation_ft
1     G_AL1         19    1.3896        Aiea       low         1042
2     G_Al2         23    1.7099        Aiea       mid         1243
3     G_AL3         20    1.1817        Aiea      high         1483
4     G_MA1         19    0.9558      Manana       low          928
5     G_MA2         22    1.6905      Manana       mid         1161
6     G_MA3         14    1.1708      Manana      high         1476
  elevation_max_diff      scientificname       commonname shortname island
1                441 Psidium cattleyanum strawberry guava     Guava   Oahu
2                441 Psidium cattleyanum strawberry guava     Guava   Oahu
3                441 Psidium cattleyanum strawberry guava     Guava   Oahu
4                548 Psidium cattleyanum strawberry guava     Guava   Oahu
5                548 Psidium cattleyanum strawberry guava     Guava   Oahu
6                548 Psidium cattleyanum strawberry guava     Guava   Oahu
  coord_x coord_y read_sum elevation_m shannon.y simpson richness.y
1   21.40  -157.9     7392       317.6    1.3896  0.6607         19
2   21.40  -157.9     2989       378.9    1.7099  0.7083         23
3   21.41  -157.9    21205       452.0    1.1817  0.4996         20
4   21.43  -157.9     7849       282.9    0.9558  0.4116         19
5   21.44  -157.9    13847       353.9    1.6905  0.7108         22
6   21.44  -157.9    35685       449.9    1.1708  0.4981         14
```


Shannon diversity along elevation
Shannon diversity
4
3
2
1
250
500
750
Elevation (m)


Richness along elevation by host
ZOTU richness
100
75
50
25
0
250
500
750
Elevation (m)

```
Call:
lm(formula = shannon ~ elevation_m, data = div_df)

Residuals:
   Min     1Q Median     3Q    Max
-1.328 -0.590 -0.282  0.669  1.615

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.644514   0.351164    7.53  1.4e-08 ***
elevation_m -0.000722   0.000609   -1.19     0.24
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.856 on 32 degrees of freedom
Multiple R-squared:  0.0421,    Adjusted R-squared:  0.0121
F-statistic: 1.41 on 1 and 32 DF,  p-value: 0.245


Analysis of Variance Table

Response: shannon
                     Df Sum Sq Mean Sq F value Pr(>F)
elevation_m           1   1.03   1.029    1.38   0.25
shortname             2   1.86   0.931    1.25   0.30
elevation_m:shortname 2   0.73   0.366    0.49   0.62
Residuals            28  20.85   0.744
```

	ienv.cp	zenv.cp	freq	maxgrp	IndVal	zscore	5%	10%	50%	90%
zotu1016	457.7	457.7	4	1	21.05	1.79	317.6	335.7	451.0	465.1
zotu1017	457.7	457.7	6	1	31.58	3.20	353.9	366.4	457.7	465.3
zotu1045	452.0	452.0	3	1	16.67	1.93	317.6	317.6	414.2	457.7
zotu1052	452.0	452.0	6	1	33.33	2.82	317.6	353.9	452.0	465.1
zotu1054	612.3	612.3	4	2	39.45	4.06	414.2	449.7	612.3	652.1
zotu1066	414.2	414.2	3	2	13.64	0.87	414.2	414.2	467.9	530.7
zotu109	590.2	590.2	3	2	37.50	5.03	515.1	554.0	609.8	994.6
zotu1125	590.2	590.2	3	2	21.67	2.34	353.9	457.7	569.5	811.9
zotu1167	449.9	449.9	3	1	21.43	2.85	353.9	353.9	366.4	451.0
zotu118	507.5	507.5	7	2	50.52	5.28	489.2	496.7	566.9	744.2
zotu12	451.0	451.0	5	1	31.25	3.35	316.7	317.6	449.9	451.0
zotu122	353.9	457.7	6	2	22.22	0.71	335.7	353.9	451.0	590.2
zotu123	507.5	507.5	5	2	32.48	3.28	414.2	414.2	529.4	744.2
zotu124	470.8	470.8	5	2	35.71	4.23	465.1	465.1	507.5	841.4

zotu1265	449.9	449.9	4	1	28.57	4.08	300.2	317.6	414.2	451.0
zotu1270	744.2	744.2	3	2	36.83	5.52	522.6	550.9	630.3	994.6
zotu129	481.0	470.8	3	2	23.08	2.46	467.9	467.9	481.0	994.6
zotu1299	457.7	457.7	3	1	15.79	1.44	414.4	449.7	451.0	457.7
zotu13	470.8	470.8	4	2	17.37	1.04	353.9	353.9	470.8	930.1
zotu144	590.2	590.2	3	2	37.50	5.56	499.4	553.2	590.2	744.2
zotu1451	457.7	457.7	5	1	26.32	2.47	353.9	366.4	451.0	465.1
zotu1452	457.7	457.7	5	2	33.33	4.29	457.7	457.7	538.3	841.4
zotu1454	457.7	470.8	6	1	31.58	2.75	353.9	366.4	457.7	465.1
zotu147	530.7	530.7	5	1	20.83	1.23	353.0	353.9	457.7	530.7
zotu1481	491.8	457.7	8	2	54.12	5.52	457.7	457.7	491.8	841.4
zotu1494	449.7	449.7	3	2	14.29	1.07	401.9	414.4	449.9	590.2
zotu15	470.8	470.8	5	2	24.32	1.88	414.2	414.2	470.8	566.9
zotu151	317.6	452.0	7	1	48.79	3.29	317.6	317.6	378.9	465.1
zotu1541	378.9	378.9	3	1	30.00	3.98	300.2	317.6	378.9	414.2
zotu1543	470.8	470.8	3	2	21.43	2.40	465.1	465.1	543.8	704.4
zotu1548	449.9	451.0	5	1	35.71	3.67	300.2	353.9	378.9	451.0
zotu1557	317.6	353.9	8	1	46.61	2.50	300.2	300.2	353.9	530.7
zotu1564	457.7	457.7	4	1	21.05	1.85	353.9	353.9	451.0	465.1
zotu159	744.2	507.5	3	2	14.87	0.26	353.9	366.4	507.5	841.4
zotu16	470.8	457.7	3	2	21.43	2.26	465.1	465.1	470.8	930.1
zotu1611	457.7	457.7	7	1	36.84	3.39	366.4	414.2	457.7	465.1
zotu168	612.3	612.3	5	2	53.66	4.91	489.1	491.8	612.3	652.1
zotu1687	507.5	507.5	3	2	27.27	3.28	477.7	496.7	507.5	930.1
zotu17	671.6	481.0	12	2	64.10	3.96	456.3	457.7	671.6	841.4
zotu18	530.7	530.7	6	1	25.00	1.36	300.2	317.6	366.4	530.7
zotu184	744.2	744.2	3	2	36.83	5.52	522.6	550.9	630.3	994.6
zotu1848	457.7	457.7	3	1	15.79	1.34	353.9	353.9	451.0	464.2
zotu19	457.7	457.7	4	2	26.67	3.02	457.7	457.7	457.7	905.9
zotu1903	378.9	449.9	5	1	36.23	4.21	353.9	353.9	414.2	451.0
zotu195	457.7	457.7	4	1	21.05	1.86	353.9	353.9	451.0	465.1
zotu1963	378.9	414.2	3	1	27.27	3.21	317.6	335.7	378.9	414.2
zotu1964	378.9	378.9	4	1	36.36	4.85	335.7	353.9	414.2	414.4
zotu198	449.9	449.9	4	1	28.57	3.64	300.2	317.6	449.7	451.0
zotu20	457.7	457.7	23	1	70.08	5.35	414.4	449.7	457.7	465.1
zotu201	451.0	451.0	7	2	28.07	2.15	335.7	409.3	457.7	994.6
zotu203	671.6	470.8	6	2	60.22	4.23	465.1	470.8	671.6	768.9
zotu2114	378.9	378.9	3	1	30.00	3.90	353.9	353.9	378.9	414.2
zotu213	744.2	744.2	3	2	36.83	4.12	353.9	496.7	671.6	808.6
zotu2146	378.9	414.2	4	1	36.36	4.40	353.9	353.9	378.9	414.2
zotu22	590.2	470.8	7	2	40.62	3.09	414.2	414.2	507.5	609.8
zotu223	470.8	470.8	7	2	38.38	3.66	414.2	414.2	470.8	553.2
zotu2238	457.7	457.7	3	1	15.79	1.58	365.7	378.9	451.0	457.7

zotu229	590.2	590.2	3	2	21.67	2.34	353.9	457.7	569.5	811.9
zotu23	353.9	353.9	4	1	57.14	7.14	300.2	300.2	353.9	366.4
zotu240	744.2	744.2	4	2	56.74	5.81	515.0	523.0	744.2	841.4
zotu241	449.9	449.9	4	1	28.57	4.19	353.9	366.4	414.4	451.0
zotu249	744.2	590.2	4	2	34.12	2.70	414.2	449.7	590.2	744.2
zotu25	671.6	457.7	13	2	62.06	3.08	366.4	366.4	566.9	671.6
zotu29	507.5	457.7	11	2	73.96	7.49	465.1	470.8	507.5	671.6
zotu294	481.0	481.0	7	1	33.33	3.09	335.7	335.7	451.0	489.1
zotu305	507.5	507.5	3	2	27.27	3.26	483.6	496.7	507.5	841.4
zotu310	507.5	507.5	3	2	27.27	3.33	489.2	496.7	507.5	684.5
zotu313	744.2	744.2	4	2	56.74	7.60	538.1	551.4	671.6	994.6
zotu317	470.8	470.8	3	1	15.00	1.04	353.9	353.9	451.0	473.5
zotu32	530.7	530.7	5	1	20.83	1.29	353.9	353.9	452.0	530.7
zotu33	414.2	414.2	3	1	25.00	3.19	300.2	317.6	378.9	414.2
zotu331	449.9	449.9	3	1	21.43	2.90	353.9	366.4	449.9	451.0
zotu34	317.6	317.6	3	1	36.83	4.98	300.2	317.6	335.7	452.0
zotu354	507.5	507.5	4	2	36.36	4.47	496.7	499.4	569.5	744.2
zotu384	414.2	414.2	3	2	13.64	0.70	353.9	414.2	470.8	671.6
zotu388	530.7	530.7	3	1	12.50	0.38	366.4	366.4	449.9	530.7
zotu406	414.2	414.2	8	1	54.12	5.50	316.7	317.6	414.2	452.0
zotu42	507.5	457.7	8	2	47.05	4.11	414.2	457.7	543.8	744.2
zotu420	449.9	449.9	3	2	15.00	1.16	400.1	414.4	451.0	507.5
zotu427	566.9	530.7	3	2	33.33	4.09	478.5	543.8	566.9	612.3
zotu438	335.7	452.0	10	1	68.63	5.50	335.7	335.7	351.1	457.7
zotu46	744.2	491.8	6	2	51.18	4.00	481.6	489.1	499.4	744.2
zotu482	481.0	481.0	7	2	53.85	6.45	467.9	473.5	489.1	841.4
zotu484	366.4	366.4	7	1	62.89	7.04	300.2	300.2	366.4	366.4
zotu485	744.2	744.2	6	2	51.18	3.55	414.2	414.2	684.9	768.9
zotu49	452.0	452.0	6	2	26.53	1.99	366.4	366.4	452.0	612.3
zotu529	317.6	457.7	7	1	48.79	2.74	300.2	300.2	353.9	457.7
zotu53	414.2	414.2	5	1	41.67	5.55	401.7	414.2	414.2	414.4
zotu56	449.7	449.7	3	2	14.29	0.91	414.4	449.7	451.0	530.7
zotu563	452.0	452.0	7	1	41.18	4.01	317.6	335.7	451.0	457.7
zotu578	744.2	744.2	5	2	53.81	4.51	457.7	457.7	704.4	768.9
zotu582	353.9	414.2	4	1	39.45	4.07	316.7	317.6	366.4	414.2
zotu60	566.9	566.9	5	2	40.77	4.05	465.1	465.1	566.9	744.2
zotu635	414.2	414.2	4	2	18.18	1.14	414.2	414.2	470.8	908.3
zotu664	530.7	530.7	6	1	25.00	1.47	353.9	353.9	452.0	530.7
zotu67	507.5	457.7	12	2	70.57	6.22	457.7	457.7	507.5	671.6
zotu681	671.6	612.3	3	2	30.11	3.20	413.8	449.9	652.1	905.9
zotu69	566.9	457.7	9	2	56.50	4.92	465.1	470.8	543.8	671.6
zotu690	744.2	507.5	3	2	14.87	0.42	414.2	414.2	507.5	841.4
zotu7	457.7	457.7	17	1	34.24	0.33	317.6	335.7	457.7	994.6

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zotu70     317.6   457.7   11      1  61.46   3.83 317.6 335.7 414.2 465.1
zotu71     451.0   451.0   13      2  41.54   2.18 335.7 353.9 452.0 744.2
zotu717    566.9   566.9    6      1  24.00   1.12 317.6 353.3 451.0 566.9
zotu72     566.9   566.9    6      2  51.82   5.89 465.1 470.8 566.9 671.6
zotu74     470.8   470.8    6      2  42.86   5.33 465.1 465.1 507.5 811.9
zotu746    366.4   366.4    3      1  18.83   1.78 353.9 353.9 366.4 457.7
zotu754    491.8   491.8    5      2  41.67   4.86 481.6 483.7 491.8 612.3
zotu770    481.0   481.0    4      2  19.13   1.28 300.2 300.2 481.0 902.2
zotu797    452.0   452.0    3      1  16.67   1.40 353.9 365.1 414.4 457.7
zotu80     744.2   457.7    7      2  48.79   2.93 414.2 414.2 631.9 744.2
zotu87     744.2   507.5    5      2  53.81   5.25 499.4 499.4 671.6 841.4
zotu9      457.7   457.7   14      1  73.68   8.26 335.7 335.7 457.7 467.9
zotu900    378.9   378.9    3      1  30.00   3.98 300.2 317.6 378.9 414.2
zotu955    317.6   449.9    4      1  34.12   2.84 300.2 300.2 353.9 449.9
zotu957    590.2   744.2    3      2  37.50   4.81 499.0 554.0 590.2 841.4
           95% purity reliability z.median filter
zotu1016  465.1  0.942       0.316    2.139      0
zotu1017  467.9  0.938       0.620    2.918      0
zotu1045  465.1  0.902       0.230    2.038      0
zotu1052  467.9  0.994       0.692    3.350      0
zotu1054  705.5  0.942       0.602    3.775      0
zotu1066  530.7  0.632       0.106    1.393      0
zotu109  1041.0  0.962       0.678    5.048      0
zotu1125  841.4  0.936       0.364    2.607      0
zotu1167  451.0  0.948       0.388    2.647      0
zotu118   744.2  0.994       0.904    5.094      0
zotu12    451.0  0.996       0.652    3.412      0
zotu122   612.3  0.378       0.322    1.975      0
zotu123   905.9  0.962       0.586    3.501      0
zotu124   841.4  0.994       0.752    4.438      0
zotu1265  451.0  0.986       0.576    3.332      0
zotu1270 1041.0  0.962       0.680    4.898      0
zotu129  1083.3  0.954       0.434    2.826      0
zotu1299  465.1  0.648       0.188    1.622      0
zotu13   1038.8  0.790       0.284    1.842      0
zotu144   747.9  0.958       0.716    4.923      0
zotu1451  465.2  0.956       0.570    2.958      0
zotu1452  841.4  0.994       0.740    4.050      0
zotu1454  467.9  0.942       0.602    2.890      0
zotu147   551.4  0.608       0.194    1.654      0
zotu1481  841.4  1.000       0.972    5.670      2
zotu1494  590.2  0.654       0.116    1.395      0
zotu15    744.2  0.914       0.442    2.482      0
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zotu151	465.1	0.998	0.752	3.675	0
zotu1541	414.4	0.956	0.566	3.667	0
zotu1543	744.2	0.966	0.436	2.812	0
zotu1548	451.0	0.994	0.778	4.078	0
zotu1557	530.7	0.986	0.670	3.271	0
zotu1564	465.1	0.932	0.368	2.341	0
zotu159	841.4	0.782	0.224	1.778	0
zotu16	1038.8	0.964	0.410	2.820	0
zotu1611	467.9	0.966	0.762	3.401	0
zotu168	682.3	0.996	0.890	5.724	0
zotu1687	994.6	0.950	0.534	3.629	0
zotu17	903.2	1.000	0.962	5.287	2
zotu18	551.4	0.938	0.442	2.365	0
zotu184	1041.0	0.962	0.680	4.898	0
zotu1848	465.1	0.828	0.184	1.653	0
zotu19	994.6	0.976	0.546	3.034	0
zotu1903	451.0	0.994	0.778	4.191	0
zotu195	465.1	0.938	0.354	2.324	0
zotu1963	414.4	0.942	0.560	3.739	0
zotu1964	414.4	0.986	0.728	4.104	0
zotu198	451.0	0.984	0.562	3.239	0
zotu20	467.9	1.000	0.996	5.538	1
zotu201	994.6	0.850	0.480	2.547	0
zotu203	994.6	0.992	0.890	5.623	0
zotu2114	414.4	0.970	0.568	3.624	0
zotu213	994.6	0.948	0.602	4.136	0
zotu2146	414.4	0.980	0.768	4.648	0
zotu22	629.8	0.988	0.774	4.097	0
zotu223	841.4	0.956	0.722	3.712	0
zotu2238	465.1	0.808	0.246	1.839	0
zotu229	841.4	0.936	0.364	2.607	0
zotu23	367.0	0.986	0.892	6.136	0
zotu240	930.1	0.990	0.838	6.011	0
zotu241	451.0	0.982	0.610	3.337	0
zotu249	744.3	0.938	0.558	3.637	0
zotu25	704.4	1.000	0.940	4.888	0
zotu29	673.3	1.000	0.990	7.302	2
zotu294	496.7	0.990	0.696	3.245	0
zotu305	841.4	0.956	0.550	3.630	0
zotu310	744.2	0.968	0.564	3.740	0
zotu313	1038.8	0.984	0.844	5.896	0
zotu317	473.5	0.772	0.104	1.463	0
zotu32	551.4	0.716	0.220	1.725	0

zotu33	414.4	0.958	0.608	3.827	0
zotu331	451.0	0.948	0.344	2.327	0
zotu34	457.7	0.920	0.432	2.903	0
zotu354	744.2	0.984	0.800	4.702	0
zotu384	671.6	0.748	0.144	1.434	0
zotu388	530.7	0.540	0.100	1.348	0
zotu406	457.7	1.000	0.960	5.534	1
zotu42	808.6	0.998	0.856	4.540	0
zotu420	507.5	0.636	0.118	1.449	0
zotu427	744.2	0.950	0.686	4.542	0
zotu438	457.7	1.000	0.964	5.712	1
zotu46	808.6	1.000	0.924	5.495	0
zotu482	930.1	1.000	0.944	5.527	0
zotu484	401.7	1.000	0.950	6.281	1
zotu485	808.6	0.994	0.686	3.752	0
zotu49	711.4	0.918	0.362	2.084	0
zotu529	465.1	0.996	0.744	3.813	0
zotu53	414.4	0.992	0.866	4.933	0
zotu56	566.9	0.750	0.118	1.446	0
zotu563	465.1	1.000	0.832	4.054	0
zotu578	808.6	0.992	0.748	4.496	0
zotu582	414.4	0.990	0.786	4.693	0
zotu60	744.2	0.998	0.816	4.568	0
zotu635	994.6	0.944	0.278	1.889	0
zotu664	551.4	0.778	0.380	2.146	0
zotu67	671.6	1.000	0.996	7.001	2
zotu681	994.6	0.900	0.408	2.806	0
zotu69	671.6	0.998	0.944	5.694	0
zotu690	841.4	0.806	0.262	1.668	0
zotu7	994.6	0.576	0.378	1.796	0
zotu70	467.9	1.000	0.986	5.728	1
zotu71	994.6	0.854	0.652	2.869	0
zotu717	569.5	0.794	0.268	1.749	0
zotu72	841.4	0.998	0.918	5.614	0
zotu74	841.4	1.000	0.896	4.883	0
zotu746	457.7	0.912	0.282	2.136	0
zotu754	930.1	0.998	0.854	4.710	0
zotu770	994.6	0.626	0.398	2.604	0
zotu797	465.1	0.918	0.208	1.905	0
zotu80	808.6	0.996	0.714	3.659	0
zotu87	930.1	0.996	0.896	5.793	0
zotu9	474.9	1.000	0.996	6.917	1
zotu900	414.4	0.956	0.566	3.667	0

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zotu955   451.0  0.984       0.630    4.185      0
zotu957   841.4  0.956       0.674    4.957      0
          cp  0.05  0.10  0.50  0.90  0.95
sumz-  457.7 378.9 414.2 456.6 465.1 467.9
sumz+  457.7 457.7 457.7 503.7 671.6 744.2
fsumz- 457.7 414.2 414.2 457.7 467.9 467.9
fsumz+ 457.7 456.4 457.7 457.7 507.5 546.2

Community CP (abundance): 507.5 m | (PA): 503.7 m

Community CP (abundance): 507.5 m | (PA): 503.7 m
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22 Summary TITAN

TITAN indicates a reproducible community turnover around ~500 m along the elevation gradient, robust to abundance vs presence/absence coding. The convergence of abundance and PA results suggests the breakpoint reflects taxon replacement rather than abundance shifts. In the context of invasion dynamics, this threshold is consistent with a decline of low-elevation assemblages (expected to include more introduced taxa) and an increase of higher-elevation assemblages. A direct enrichment test (direction $\times$ invasion status) can formalize this, but the community-level signal already supports an elevation-linked transition.